

Actuation-Envelope-Constrained Centroidal NMPC for On-Orbit Assembly: Multi-Step Locomotion of a Dual-Arm Robot on a Free-Floating Structure

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ABSTRACT A robot crawling across a free-floating orbital structure exchanges angular momentum with the host at every step, while the host attitude-control system has finite reaction torque and finite stored-momentum capacity. We formulate this interaction as an *allocated locomotion envelope*: a centroidal nonlinear model-predictive controller (NMPC) generates online locomotion references whose planned robot-on-structure reaction is bounded at feasible single-support discretisation knots and whose centroidal state is constrained by a locally refreshed reduced prediction of reaction-wheel momentum. A whole-body quadratic program executes the plan at high rate under the full multibody, contact and joint constraints, while the attitude-control system (AOCS) remains dynamically separate; measured wheel momentum enters the locomotion optimisation only through frozen per-cycle parameters. In a six-step simulated traversal at a 1% robot-to-structure mass ratio, all six docking transitions satisfy the implemented HOTDOCK-inspired capture gate, the planned reaction reaches but does not exceed the 2.5 N·m per-axis allocation at feasible single-support knots, the measured wheel momentum peaks at 4.10 N·m·s (82% of the allocated box), and the platform attitude remains below 0.37° per axis during the traversal. At this operating point the storage constraint is inactive; consequently the reported experiment establishes torque-envelope shaping by the predictive planner, but does not by itself establish a benefit of prediction horizon over an otherwise equivalent horizon-free controller. The executed AOCS command can also saturate because attitude-feedback demand is not reserved inside the planner allocation; this distinction is quantified and retained explicitly in the claims.

INDEX TERMS actuation envelope, centroidal dynamics, free-floating space robot, in-orbit assembly, nonlinear model-predictive control, reaction wheel assembly, whole-body control.

I. INTRODUCTION

ON-ORBIT assembly of large structures such as telescopes, antennas and solar arrays calls for robots that relocate themselves across the structure they are constructing. A crawling robot that anchors, swings an arm to the next interface and docks exchanges linear and angular momentum with the free-floating host. The host attitude-control system must absorb the resulting rotational disturbance with momentum-exchange actuators whose instantaneous torque and stored angular momentum are finite. A locomotion plan can therefore be kinematically feasible and executable by the robot while being incompatible with the authority available to the host attitude system.

This work addresses that interface at the motion-generation

level. A centroidal nonlinear model-predictive controller (NMPC) generates locomotion references online while constraining two quantities derived from angular momentum balance about the structure centre of mass: (i) the planned robot-on-structure reaction moment and (ii) a locally refreshed reduced prediction of reaction-wheel spin momentum. A high-rate whole-body quadratic program (QP) executes the resulting centroidal and end-effector references under the full multibody, bilateral-contact and joint-actuation constraints. The reaction-wheel dynamics are not co-integrated into either optimisation; Stage 1 receives a frozen reduced-momentum offset refreshed from the measured wheel momentum, and the whole-body layer receives the current measured wheel momentum for a soft one-step safeguard. The AOCS is otherwise

an independent feedforward-plus-feedback subsystem.

The distinction between *planned robot-induced reaction* and *total AOCS command* is essential. The NMPC constrains the former at feasible single-support discretisation knots. It does not reserve torque for the AOCS feedback terms, does not provide a by-construction guarantee between knots or during the inter-step double-support phase, and does not guarantee recursive feasibility of the nonlinear programme. In the present simulation the numerical planning allocation and the physical wheel-command clamp are both 2.5 N·m per axis; because the feedback reserve is set to zero, the total pre-clipping AOCS command can exceed that value even when the planned robot reaction satisfies its bound. These scopes are kept explicit throughout the paper rather than folded into a single “hard actuator constraint” claim.

The six-step experiment studied here is a *torque-limited* operating point. At a 1% robot-to-structure mass ratio, all six docking transitions complete within the implemented HOTDOCK-inspired capture gate. The planned reaction reaches the 2.5 N·m per-axis allocation without exceeding it at feasible single-support NMPC knots, the measured wheel momentum peaks at 4.10 N·m·s (82% of the allocated 5 N·m·s box), and the structure attitude remains below 0.37° per axis during the traversal. The storage inequality is present but inactive. Consequently, this experiment demonstrates that the NMPC reshapes the planned reaction demand; it does *not* establish that a finite prediction horizon is necessary at this operating point. A storage-active case and an otherwise equivalent horizon-free baseline are required for that stronger conclusion.

The contributions supported by the reported data are therefore stated narrowly:

- 1) a conservation-based reduction that maps a host reaction-torque allocation and a measured wheel-momentum offset into constraints expressed only in robot-centroidal variables, without carrying the AOCS state in the locomotion NMPC;
- 2) a two-stage centroidal-NMPC/whole-body-QP architecture for multi-step crawling on a free-floating host, with an explicit contract between the planning layer, the executed contact wrench and the separately regulated reaction-wheel system; and
- 3) a closed-loop six-step simulation that quantifies docking, reaction-envelope activity, wheel momentum, attitude, whole-body tracking, solver cost and the limitations of the present validation, including the inactivity of the storage bound and the absence of an AOCS feedback reserve.

The 2.5 N·m and 5 N·m·s boxes used below are allocated demonstration envelopes for the orthogonal three-wheel abstraction, not the sizing result of a particular flight unit. Mapping the same planner interface to a redundant reaction-wheel or control-moment-gyroscope cluster requires a separate proof that the allocated set is contained in the jointly feasible hardware envelope, including steering and singularity constraints where applicable (Section VIII).

II. RELATED WORK

Nakamura and Mukherjee [1] addressed momentum avoidance for free-flying robots via nonholonomic path planning, without RWA storage limits as explicit real-time constraints. An orthogonal line renders the motion momentum-neutral by planning in the reaction null space of the base coupling [2], restricting the achievable motions; we instead admit momentum exchange and bound it within the actuator envelope (Section VIII).

Model-predictive momentum management is also being developed directly for spacecraft attitude systems. Shen and Caverly use MPC to coordinate solar-sail momentum-management actuators while enforcing actuator constraints, first for mass-translation and reflectivity devices [3] and subsequently with disturbance-torque estimation and an explicit four-reaction-wheel assembly [4]. Those studies predict the spacecraft and its momentum-management actuators as the controlled plant. The problem considered here is different: the crawler is a mechanically distinct disturbance source, and the locomotion optimiser deliberately does not integrate the host AOCS dynamics. The novelty claimed here is therefore not “MPC for momentum management” in general, but the reduced host-robot actuation-allocation interface embedded in an online centroidal locomotion planner.

A substantial body of work controls the free-floating *manipulator* itself: NMPC realises end-effector trajectories on free-floating servicers [5], convex programs generate kinematically feasible paths [6], and trajectory optimisation handles free-floating two-arm systems [7]. Coordinated schemes go further and allocate the base actuators explicitly: thrusters, reaction wheels and arm in [8]; attitude-prioritised null-space reconfiguration with reaction wheels in [9]. In all of these the actuated base is the robot’s own; ours carries none, and the momentum it exchanges is absorbed by the attitude system of a *separate* host structure, which the locomotion layer treats as a bounded capacity rather than a co-controlled plant.

Closest to our work is Lutze et al. [10], who study the MIRROR multi-arm robot with HOTDOCK interfaces and address the same problem: reducing the attitude disturbance a walking multi-arm robot imparts to a free-floating structure. They derive a closely related conservation relation through the $SE(3)$ adjoints and likewise cite reaction-wheel saturation, without modelling a specific actuator; their controller is a reactive QP minimising the base rotation as a *soft*, weighted cost, with the robot treated as a point mass in continuously moving contact and no attitude actuator on the platform. Three differences distinguish our approach. First, we impose the disturbance bound as a *hard* constraint on the attitude-actuation envelope, so the feasible single-support knot-wise plan lies within the allocated rate bound and the reduced predicted-storage box by construction. Second, the constraint is embedded in a receding-horizon centroidal NMPC generating the reference online, rather than a reactive QP tracking a pre-computed trajectory. Third, it is parameterised by the allocated envelope and a reduced-momentum offset refreshed from measurements, keeping the AOCS dynamics and actu-

ator state outside the locomotion optimisation. As in their work, we do not render the gait angular-momentum-neutral.

Most directly related in mission and platform is the co-design framework of Rognant *et al.* [11], likewise targeting on-orbit assembly of large modular structures with a crawling robot carrying two VISPA arms [12]. Building on the momentum-conservation methodology of [13], they bound the robot's reachable centre-of-mass velocity from the actuator capacity (their Eq. 6) and derive the corresponding minimum step time, which serves as the cost function of a symbolic assembly-sequence planner checked by finite-element analysis. Their actuator bound serves an offline plan-ranking cost, folds the torso- and joint-angular-momentum terms of their full constraint (Eq. 5) into a fixed motion-pattern assumption, and is assessed through planning benchmarks. We instead retain the centroidal angular momentum L_{com} as an NMPC state, constrain both the actuator torque and the stored momentum per axis online, and report closed-loop locomotion in a multibody simulator together with the resulting platform attitude. The two approaches are complementary: their planner selects an assembly sequence under an a-priori actuator-capacity estimate, while our scheme provides the closed-loop motion generation that enforces that capacity during each step. Table 1 summarises the positioning.

A recent line of trajectory-optimisation work addresses multi-arm robots that dock to and move across a target spacecraft: four-arm optimisation with contact timing, body motion, and interaction forces as decision variables, tracked by a weighted convex controller [14]; hybrid task-and-motion planning for on-orbit loco-manipulation [15]; and, closest in architecture, offline contact-aware trajectory optimisation with reinforcement-learning tracking for a thruster-equipped free-flying multi-arm robot [16]. These works optimise *how the robot moves* — contacts, timing, body actuation; none of the representative works reviewed here imposes the host's finite reaction-torque and momentum-storage allocation as an online hard constraint in a centroidal locomotion planner.

III. MISSION CONTEXT AND SYSTEM DESCRIPTION

A. DEPLOYMENT SCENARIO

The scenario considered is the incremental robotic assembly of a large space-based structure, as studied in the companion planning work [11]. The reference orbit is geostationary. The system comprises a free-floating host spacecraft equipped with a reaction wheel assembly (RWA); CFRP lattice truss beams with HOTDOCK [19] connectors; and a dual-arm crawling robot whose 7-DoF VISPA-class arms [12] serve simultaneously as manipulators and locomotion legs. Each end-effector is HOTDOCK-compatible, enabling locomotion by alternately grasping structural anchors. Simulation parameters are listed in Table 3.

B. LOCOMOTION PHASES

Each locomotion step cycles through two controller phases (Fig. 2):

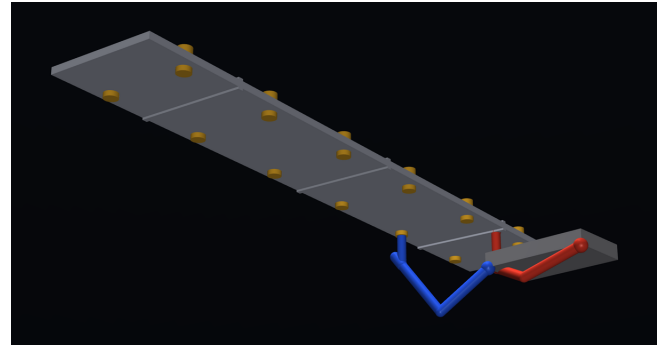


FIGURE 1. Mission scenario. A dual-arm robot (71 kg, 2×7 DoF) crawls across a free-floating assembly structure (7.1 t) whose attitude is held by on-board momentum-exchange actuation; the docking interfaces form the locomotion path. Every step exchanges momentum with the platform, and the locomotion plan is constrained to keep that exchange within the actuation envelope.

- **Double Support (DS):** both arms anchored to the structure and the torso held near the docked configuration by the whole-body QP. The centroidal NMPC is bypassed during the inter-step transition and remains applied, on the frozen contact set, during the terminal hold (Section VI-A); the reaction wheels remain commanded at every control cycle (Section VI-A).
- **Single Support (SS):** stance arm anchored, swing arm driven along a quintic trajectory toward the next anchor by the centroidal NMPC and whole-body QP; the phase ends when the implemented three-condition capture gate is satisfied—position, orientation and relative-twist thresholds, defined below—triggering the next DS phase.

Over an assembly sequence spanning tens of metres, the cumulative momentum exchanged between robot and spacecraft during successive SS phases motivates the predictive constraint enforcement developed in Sections V–VI.

IV. KINEMATICS, DYNAMICS, AND MOMENTUM BALANCE

A. GENERALISED COORDINATES

The coupled system (spacecraft + robot) is described by four groups of generalised coordinates:

- **Spacecraft pose:** $\mathbf{q}_s = [\mathbf{p}_s^\top, \mathbf{r}_s^\top]^\top \in \mathbb{S}^3 \times \mathbb{R}^3$, where $\mathbf{p}_s \in \mathbb{S}^3 \subset \mathbb{H}$ is a unit quaternion and $\mathbf{r}_s \in \mathbb{R}^3$ is the CoM position in the inertial frame $\mathcal{R}_{\text{ine}}$.
- **Reaction wheel angles:** $\mathbf{q}_w \in \mathbb{R}^{n_w}$; the stored angular momentum is $\mathbf{h}_w = \mathbf{B}_w \dot{\mathbf{q}}_w \in \mathbb{R}^3$, with $\mathbf{B}_w \in \mathbb{R}^{3 \times n_w}$ the wheel-momentum allocation map (defined in (IV-C.1)).
- **Robot torso pose:** $\mathbf{q}_t = [\mathbf{p}_t^\top, \mathbf{r}_t^\top]^\top \in \mathbb{S}^3 \times \mathbb{R}^3$, with $\mathbf{p}_t$ likewise a unit quaternion.
- **Active joint angles:** $\mathbf{q} \in \mathbb{R}^{n_q}$, with $n_q = \sum_{i=1}^m n_i$ over m kinematic chains.

For compactness the platform and robot generalised coordinates are grouped as $\mathbf{q}_p = [\mathbf{q}_s^\top, \mathbf{q}_w^\top]^\top$ and $\mathbf{q}_r = [\mathbf{q}_t^\top, \mathbf{q}^\top]^\top$, so that $\ddot{\mathbf{q}}_p$ and $\ddot{\mathbf{q}}_r$ are the acceleration vectors appearing directly in the KKT system (IV-C.4). Because the poses live

TABLE 1. Positioning of the proposed approach against related space-robotics work. The comparison is scoped to approaches addressing the disturbance a multi-arm robot imparts to its supporting structure; entries are representative, not exhaustive.

	Mishra '22 [17]	Kalaycioglu '23 [18]	Rognant '19,'25 [11], [13]	Lutze '23 [10]	Belmonte '26 [16]	This work
Task	Loco + manip.	Manipulation	Loco + planning	Locomotion	Loco + manip.	Locomotion
Predictive framework	— (passivity)	PNMPC	— (offline)	— (reactive QP)	TO (offline) + RL	Centroidal NMPC
Attitude-actuator envelope	—	—	Open-loop (Eq. 6)	—	— (body thrusters)	Planned SS-knot τ + reduced h
Actuator-aware without AOCS co-integration	—	—	—	—	—	Yes
Contact model	Walking	Grasp	Geometric steps	Continuous pt.	Optimised contacts	Multi-step + DS/SS
Closed-loop validation	Sim	Sim	Planner + FEM	Sim	Sim (RL tracking)	Sim (6-step, 1%)

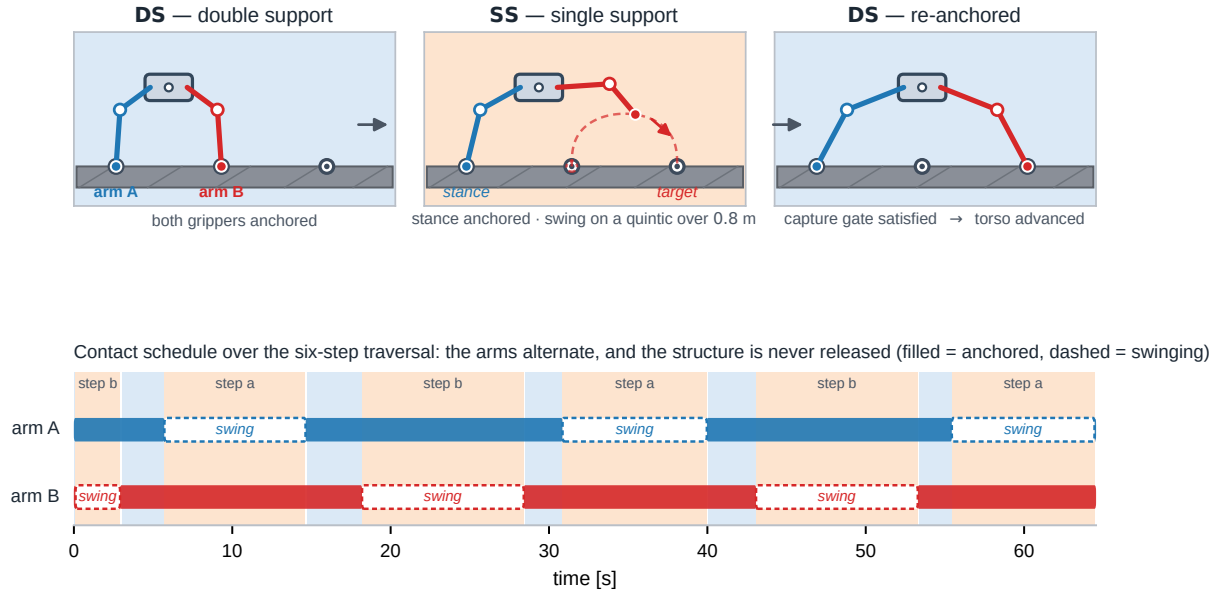


FIGURE 2. The two-phase locomotion cycle. In **DS** both grippers are anchored; in **SS** the stance arm stays anchored while the swing arm follows a quintic over the 0.8 m step, until the capture gate is satisfied: gripper-to-anchor distance below 5 mm, orientation error below 5° , and length-normalised twist $\xi_{\text{cap}} = \|\mathbf{v}_{\text{rel}}^T, \ell_c \omega_{\text{rel}}^T\|_2 < 0.05$ m/s with $\ell_c = 1$ m — an exact rewriting of the implemented test, inspired by the HOTDOCK interface [19] rather than derived from its qualification data (values at capture: Table 4). **Bottom:** contact schedule of the six-step traversal; at least one gripper is anchored at every instant. Phase colours match the time-series figures.

on a manifold, their velocities are twists rather than componentwise derivatives of the seven stored numbers:

$$\dot{\mathbf{q}}_s = \begin{bmatrix} \boldsymbol{\omega}_s \\ \dot{\mathbf{r}}_s \end{bmatrix} \in \mathbb{R}^6, \quad \dot{\mathbf{q}}_t = \begin{bmatrix} \boldsymbol{\omega}_t \\ \dot{\mathbf{r}}_t \end{bmatrix} \in \mathbb{R}^6, \quad (\text{IV-A.0})$$

each stacking an angular and a linear velocity in the frame convention of (IV-A.1); every Jacobian below acts on these twists, never on a raw pose derivative. Accordingly the robot generalised velocity $\dot{\mathbf{q}}_r$ has dimension $n_v = 6 + n_q$ (six for the floating torso, not seven), and $\ddot{\mathbf{q}}_r$ is its tangent acceleration; with the seven-joint arms used here, $n_q = 14$ and $n_v = 20$. All robot body twists are expressed in the spacecraft frame $\mathcal{R}_s$ via the Adjoint map:

$$\text{Ad}({}^a\mathbf{T}_b) = \begin{bmatrix} {}^a\mathbf{R}_b & \mathbf{0} \\ [{}^a\mathbf{r}_b]_\times {}^a\mathbf{R}_b & {}^a\mathbf{R}_b \end{bmatrix} \in \mathbb{R}^{6 \times 6}, \quad (\text{IV-A.1})$$

where ${}^a\mathbf{T}_b \in SE(3)$ is the homogeneous transform from frame b to frame a , ${}^a\mathbf{R}_b \in SO(3)$ its rotation, ${}^a\mathbf{r}_b \in \mathbb{R}^3$ its translation, and $[\cdot]_\times$ denotes the skew-symmetric matrix of the cross product. Unless a left-superscript indicates otherwise, all vectors in the sequel are expressed in the structure frame $\mathcal{R}_s$. A superposed dot denotes the derivative of those coordinates, $\dot{\mathbf{a}} = d(\mathbf{a})/dt$ taken in $\mathcal{R}_s$; the inertial derivative represented in the same frame carries the transport term, $(d\mathbf{a}/dt)_I = \dot{\mathbf{a}} + \boldsymbol{\omega}_s \times \mathbf{a}$, and is written with the explicit $(\cdot)_I$ subscript wherever it is meant. Where a transport term is dropped, the approximation is stated. The Adjoint map yields the twist of the i -th body in chain j as

$$\mathbf{v}_i^j = \text{Ad}({}^s\mathbf{T}_t) \mathbf{J}_t^j \dot{\mathbf{q}}_t + \text{Ad}({}^s\mathbf{T}_t) \mathbf{J}_q^j \dot{\mathbf{q}}^j = \tilde{\mathbf{J}}_t^j \dot{\mathbf{q}}_t + \tilde{\mathbf{J}}_q^j \dot{\mathbf{q}}^j. \quad (\text{IV-A.2})$$

The tilde is defined by this reprojection and carries it through-

out: $\tilde{\mathbf{J}}^j := \text{Ad}({}^s\mathbf{T}_t) \mathbf{J}^j$ is a Jacobian already expressed in $\mathcal{R}_s$, whereas an untilded $\mathbf{J}$ is taken in its own body frame.

B. BILATERAL CONTACT CONSTRAINT

Unlike terrestrial legged robots, where contact is unilateral, docking to a free-floating structure imposes a *bilateral* holonomic constraint: the end-effector pose must coincide with the anchor pose at all times. Differentiating this coincidence condition yields the velocity-level constraint for the j -th arm docked at anchor c_j :

$$-\mathbf{J}_s^{c_j} \dot{\mathbf{q}}_s + \tilde{\mathbf{J}}_q^j \dot{\mathbf{q}}^j + \tilde{\mathbf{J}}_t^j \dot{\mathbf{q}}_t = \mathbf{0}. \quad (\text{IV-B.1})$$

The contact wrench λ appears as a Lagrange multiplier enforcing (IV-B.1) in the dynamics.

C. COUPLED LAGRANGIAN DYNAMICS

Applying the Euler–Lagrange formalism to the full system and enforcing (IV-B.1), the dynamics partition naturally into a *platform* subsystem (spacecraft + wheels) and a *robot* subsystem (torso + joints), coupled through the contact wrench λ .

The platform inertia matrix and input vector are:

$$\mathbf{H}_p = \begin{bmatrix} \mathbf{M}_s & \mathbf{H}_{pw} \\ \star & \mathbf{H}_w \end{bmatrix}, \quad \tau_p = \begin{bmatrix} \mathbf{0} \\ \tau_w \end{bmatrix}, \quad (\text{IV-C.1})$$

where $\mathbf{M}_s$ is the spacecraft inertia matrix, $\mathbf{H}_{pw}$ the spacecraft–wheel inertial coupling, $\mathbf{H}_w$ the wheel inertia matrix, and τ_w the wheel torques (the spacecraft body carries no direct actuator, hence the zero). The allocation map of Section IV-A follows from this coupling block, $\mathbf{B}_w := \Pi_\omega \mathbf{H}_{pw}$, where Π_ω selects the angular rows.

The robot inertia matrix and input vector are:

$$\mathbf{H}_r = \begin{bmatrix} \tilde{\mathbf{H}}_t & \tilde{\mathbf{H}}_{tq} \\ \star & \mathbf{H}_q \end{bmatrix}, \quad \tau_r = \begin{bmatrix} \mathbf{0} \\ \tau_q \end{bmatrix}, \quad (\text{IV-C.2})$$

where $\tilde{\mathbf{H}}_t \in \mathbb{R}^{6 \times 6}$ and $\tilde{\mathbf{H}}_{tq} \in \mathbb{R}^{6 \times n_q}$ are the configuration-dependent composite inertia blocks of the torso and joints reprojected into $\mathcal{R}_s$, $\mathbf{H}_q$ the joint inertia matrix, and τ_q the joint torques (the torso is unactuated, hence the zero).

The contact Jacobians map generalised velocities to the constraint:

$$\mathbf{J}_p = [-\mathbf{J}_s^c \quad \mathbf{0}], \quad \mathbf{J}_r = [\tilde{\mathbf{J}}^t \quad \tilde{\mathbf{J}}^q], \quad (\text{IV-C.3})$$

where $\mathbf{J}_s^c$, $\tilde{\mathbf{J}}^t$ and $\tilde{\mathbf{J}}^q$ stack the per-arm blocks $\mathbf{J}_s^{c_j}$, $\tilde{\mathbf{J}}_t^j$, $\tilde{\mathbf{J}}_q^j$ of (IV-B.1) over the docked arms: the spacecraft contact Jacobian, and the torso and joint contact Jacobians expressed in $\mathcal{R}_s$. The zero block in $\mathbf{J}_p$ reflects that the reaction wheels are not kinematically linked to the contact constraint; they enter the coupled dynamics exclusively through the inertial coupling $\mathbf{H}_{pw}$.

Assembling these blocks, with $\mathbf{c}_p$, $\mathbf{c}_r$ collecting the quadratic velocity (Coriolis/centrifugal) terms and γ_c the constraint-acceleration bias, yields the augmented KKT system:

$$\begin{bmatrix} \mathbf{H}_p & \mathbf{0} & -\mathbf{J}_p^\top \\ \mathbf{0} & \mathbf{H}_r & -\mathbf{J}_r^\top \\ \mathbf{J}_p & \mathbf{J}_r & \mathbf{0} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{q}}_p \\ \ddot{\mathbf{q}}_r \\ \lambda \end{bmatrix} + \begin{bmatrix} \mathbf{c}_p \\ \mathbf{c}_r \\ \gamma_c \end{bmatrix} = \begin{bmatrix} \tau_p \\ \tau_r \\ \mathbf{0} \end{bmatrix}. \quad (\text{IV-C.4})$$

The multiplier λ is the contact wrench exerted *by the structure on the robot*, the convention used throughout; the off-diagonal signs follow from it, giving $\mathbf{H}_r \ddot{\mathbf{q}}_r + \mathbf{c}_r = \tau_r + \mathbf{J}_r^\top \lambda$ for the robot and the opposite action on the platform through the sign carried by $\mathbf{J}_p$.

D. CENTROIDAL REDUCTION

Solving the full system (IV-C.4) directly over a planning horizon is incompatible with the 10 Hz planning budget of the present implementation: for $n_q = 14$, $n_c = 2$ contacts, and a horizon of $N = 8$ steps, the whole-body state has ≈ 35 components and the resulting NLP involves several hundred nonlinear constraints. Following [20], [21], we therefore reduce the robot subsystem to a low-dimensional *centroidal* model, which captures the quantities relevant to momentum management while remaining tractable for online 10 Hz receding-horizon optimisation in the reported implementation. The reduction projects the dynamics onto three aggregate quantities expressed in $\mathcal{R}_s$: the centre-of-mass position $\mathbf{r}_{\text{com}} \in \mathbb{R}^3$, taken from the *structure centre of mass*; the velocity $\mathbf{v}_{\text{com}} \in \mathbb{R}^3$, the inertial velocity of the robot centre of mass represented in $\mathcal{R}_s$; and the centroidal angular momentum $\mathbf{L}_{\text{com}} \in \mathbb{R}^3$ about the *robot* centre of mass, also represented in $\mathcal{R}_s$. The first two satisfy $(d\mathbf{r}_{\text{com}}/dt)_I = \mathbf{v}_{\text{com}} - \mathbf{v}_s$, and Stage 1 propagates $\dot{\mathbf{r}}_{\text{com}} = \mathbf{v}_{\text{com}}$ under the same quasi-inertial treatment of $\mathcal{R}_s$. These are related to the robot generalised coordinates by:

$$\mathbf{r}_{\text{com}} = \frac{1}{m} \sum_b m_b \mathbf{r}_b, \quad \begin{bmatrix} m\mathbf{v}_{\text{com}} \\ \mathbf{L}_{\text{com}} \end{bmatrix} = \mathbf{A}_G(\mathbf{q}_r) \dot{\mathbf{q}}_r. \quad (\text{IV-D.1})$$

where the sum runs over all robot bodies b of mass m_b , and $\mathbf{A}_G(\mathbf{q}_r) \in \mathbb{R}^{6 \times n_r}$ is the *centroidal momentum matrix* (CMM) [20], mapping the robot generalised velocities to its aggregate linear and angular momentum reduced at the robot centre of mass; its lower block yields the centroidal angular momentum $\mathbf{L}_{\text{com}}$. This is the quantity returned by the composite-rigid-body computation used in the implementation. Differentiating (IV-D.1) and substituting the robot block of (IV-C.4) yields the centroidal dynamics:

$$\left(\frac{d(m\mathbf{v}_{\text{com}})}{dt} \right)_I = \sum_{j=1}^{n_c} \mathbf{f}_j, \quad (\text{IV-D.2a})$$

$$\left(\frac{d\mathbf{L}_{\text{com}}}{dt} \right)_I = \sum_{j=1}^{n_c} [(\mathbf{r}_{c_j} - \mathbf{r}_{\text{com}}) \times \mathbf{f}_j + \tau_j], \quad (\text{IV-D.2b})$$

written with the inertial derivative of Section IV-A. Stage 1 propagates these over a 0.8 s horizon in $\mathcal{R}_s$ and treats that frame as locally quasi-inertial. The linear balance therefore drops the force-dimensional transport $-\omega_s \times m\mathbf{v}_{\text{com}}$. The moment balance about the moving structure-CoM origin drops $-\omega_s \times \mathbf{L}_{\text{robot}}$ and $\mathbf{v}_s \times m\mathbf{v}_{\text{com}}$. Correspondingly, the position kinematics is propagated as $\dot{\mathbf{r}}_{\text{com}} \approx \mathbf{v}_{\text{com}}$, dropping $-\mathbf{v}_s$ and $-\omega_s \times \mathbf{r}_{\text{com}}$. For the moment balance, the

last two residuals are evaluated with dimensionally matched normalisation by the transmitted contact-wrench moment: the rotational term has a median ratio of 1.5×10^{-4} and a maximum of 8.4×10^{-3} , while the moving-origin term has a median of 3.5×10^{-6} and a maximum of 5.5×10^{-3} . Neither exceeds one percent. The force-dimensional linear-transport term is not included in these moment ratios and is not separately quantified here. Here $n_c \in \{1, 2\}$ is the number of active contacts (SS or DS), $\mathbf{f}_j$ and $\boldsymbol{\tau}_j$ are the force and torque of the j -th contact wrench $\boldsymbol{\lambda}_j$, and $\mathbf{r}_{C_j}$ is the j -th anchor position measured from the *structure CoM* (origin of $\mathcal{R}_s$). Note that (IV-D.2a)–(IV-D.2b) depend only on the contact wrenches $\boldsymbol{\lambda}$ [20]: joint motion enters the momentum through $\mathbf{A}_G(\mathbf{q})\dot{\mathbf{q}}$ but not its rate, which is what Stage 1 predicts and constrains (Section VI).

Combined with the wheel momentum relation (V-A.5), the centroidal state $\mathbf{x}_c = [\mathbf{r}_{\text{com}}^\top, \mathbf{v}_{\text{com}}^\top, \mathbf{L}_{\text{com}}^\top]^\top \in \mathbb{R}^9$ determines the reduced storage prediction $\mathbf{h}_w$ through the conservation law (V-C.2) and its per-cycle reduction (V-C.3), with the wheel state entering only through the frozen parameter vector $\mathbf{c}_{\text{simple}} \in \mathbb{R}^3$. This 9-dimensional centroidal state is the planning variable of the NMPC of Section VI.

V. MOMENTUM TRANSFER BOUNDS AND SATURATION CONSTRAINTS

This section derives explicit constraints on robot motion arising from the finite momentum capacity of the spacecraft actuation system. These constraints motivate the predictive control architecture of Section VI.

A. MOMENTUM CONSERVATION AND WHEEL SATURATION CONDITION

External torques are neglected, as is standard for locomotion-timescale analyses of free-floating manipulators [17], [22]: at the geostationary reference orbit the dominant gravity-gradient torque ($\sim 10^{-5}$ N·m here) contributes below 0.01% of the momentum box over a 14 s step, and below 0.3% even for structure inertias two orders of magnitude larger; its accumulation over longer windows is an attitude-system sizing question outside the present scope. With external torques neglected the total angular momentum is conserved inertially; expressed in $\mathcal{R}_s$ its coordinates are not constant but satisfy $\dot{\mathbf{c}} = -\boldsymbol{\omega}_s \times \mathbf{c}$, a transport the reduced planner neglects over one short horizon under the quasi-stationary approximation:

$$\mathbf{L}_s + \mathbf{L}_{\text{robot}} + \mathbf{L}_{\text{AOCS}} = \mathbf{c}, \quad (\text{V-A.1})$$

where $\mathbf{c} \in \mathbb{R}^3$ is the total angular momentum, an inertial constant set by the initial conditions, here written in its $\mathcal{R}_s$ coordinates. Reducing it about the *moving* structure centre of mass is legitimate because the coupled system starts from rest: the total linear momentum $\mathbf{P}_{\text{tot}}$ is zero at $t = 0$ and, with no net external force, remains zero, so no $\mathbf{v}_s \times \mathbf{P}_{\text{tot}}$ term enters the reduction. Differentiating those coordinates carries the transport term of the frame convention:

$$\dot{\mathbf{L}}_s + \dot{\mathbf{L}}_{\text{robot}} + \dot{\mathbf{L}}_{\text{AOCS}} = -\boldsymbol{\omega}_s \times \mathbf{c}, \quad (\text{V-A.2})$$

which reduces to zero under the quasi-stationary approximation $\boldsymbol{\omega}_s \approx \mathbf{0}$ that the planner adopts over one horizon, and whose residual is measured in Section VII. The storage variable used by the planner is the *relative spin* momentum of the wheel set, $\mathbf{h}_w = \mathbf{B}_w \dot{\mathbf{q}}_w$, expressed in $\mathcal{R}_s$. Its coordinate derivative and inertial derivative must be distinguished. For fixed wheel axes,

$$\left(\frac{d\mathbf{h}_w}{dt} \right)_I^s = \dot{\mathbf{h}}_w + \boldsymbol{\omega}_s \times \mathbf{h}_w, \quad \dot{\mathbf{h}}_w = \mathbf{B}_w \ddot{\mathbf{q}}_w, \quad (\text{V-A.3})$$

where the transverse wheel inertia associated with the common spacecraft rotation is included in the platform inertia rather than in $\mathbf{h}_w$. The ideal spin-axis motor command is therefore $\boldsymbol{\tau}_m = \dot{\mathbf{h}}_w$ in structure-frame coordinates, whereas the inertial change of the stored spin momentum also contains the gyroscopic transport term $\boldsymbol{\omega}_s \times \mathbf{h}_w$. The equal-and-opposite reaction transmitted through the wheel supports consequently differs from $-\boldsymbol{\tau}_m$ by this transport contribution. The reduced planner neglects that term under the same low-rate approximation used elsewhere; in the canonical trace its largest component is 1.81×10^{-3} N·m, 7.3×10^{-4} of the 2.5 N·m planning allocation.

For a reaction-wheel assembly (RWA), the commanded spin torque is thus

$$\boldsymbol{\tau}_m = \dot{\mathbf{h}}_w = \mathbf{B}_w \ddot{\mathbf{q}}_w \quad (\text{fixed-axis RWA}), \quad (\text{V-A.4})$$

while a control-moment-gyroscope cluster requires a different steering map and a configuration-dependent feasible torque–momentum set. The simulated plant uses only the fixed-axis RWA abstraction.

Since uncontrolled host attitude drift is inadmissible during precision assembly, the wheel controller is commanded to absorb the dominant robot-induced exchange. After neglecting the structure-momentum change, frame transport, the small wheel gyroscopic transport above, and external torque over one short horizon, the reduced relation used by the planner is

$$\boldsymbol{\tau}_m = \dot{\mathbf{h}}_w \approx -\dot{\mathbf{L}}_{\text{robot}}. \quad (\text{V-A.5})$$

This is an allocation model, not an identity for the total AOCS command: the executed command also contains attitude-feedback terms, and the realised robot–host exchange can differ from the NMPC plan. Those discrepancies are reported explicitly in Section VII.

Two saturation conditions follow directly from (V-A.5): the instantaneous torque demand must not exceed the wheel torque capacity, $\|\boldsymbol{\tau}_w\|_\infty \leq \tau_{w,\text{max}}$, and the accumulated stored momentum must remain within the storage limit, $\|\mathbf{h}_w\|_\infty \leq h_{\text{max}}$, where $\mathbf{h}_w = \mathbf{B}_w \dot{\mathbf{q}}_w \in \mathbb{R}^3$ (introduced in Section IV-A). Section V translates these two conditions into explicit per-axis bounds on the robot's centroidal motion, which are then enforced as hard path constraints in the NMPC of Section VI.

B. MOMENTUM COUPLING AND FEASIBLE CONTROL SET

The angular momentum of the robot about the structure CoM (the origin of $\mathcal{R}_s$) follows from the centroidal momen-

tum (IV-D.1) by the parallel-axis (König) transport of the linear momentum through the CoM lever $\mathbf{r}_{\text{com}}$:

$$\mathbf{L}_{\text{robot}}(\mathbf{q}_r, \dot{\mathbf{q}}_r) = \mathbf{L}_{\text{com}} + \mathbf{r}_{\text{com}} \times m\mathbf{v}_{\text{com}}. \quad (\text{V-B.1})$$

The moment the anchoring wrenches exert on the robot about the structure CoM is, by definition, the contact-wrench moment

$$\mathbf{M}_{s \rightarrow r} = \sum_{j=1}^{n_c} (\mathbf{r}_{C_j} \times \mathbf{f}_j + \boldsymbol{\tau}_j), \quad \boldsymbol{\tau}_{r \rightarrow s} = -\mathbf{M}_{s \rightarrow r}, \quad (\text{V-B.2})$$

whose negative $\boldsymbol{\tau}_{r \rightarrow s}$ is the physical reaction transmitted to the structure. Differentiating (V-B.1) about the moving structure-CoM origin, with coordinates in the rotating frame $\mathcal{R}_s$, relates this moment *exactly* to the momentum rate:

$$\mathbf{M}_{s \rightarrow r} = \dot{\mathbf{L}}_{\text{robot}} + \boldsymbol{\omega}_s \times \mathbf{L}_{\text{robot}} + \mathbf{v}_s \times m\mathbf{v}_{\text{com}}, \quad (\text{V-B.2b})$$

where $\dot{\mathbf{L}}_{\text{robot}}$ is the $\mathcal{R}_s$ -coordinate rate: the moving origin contributes the last term through $(d\mathbf{r}_{\text{com}}/dt)_I = \mathbf{v}_{\text{com}} - \mathbf{v}_s$, and the rotating coordinates the transport term. Stage 1 adopts the locally quasi-inertial approximation of Section IV-D and neglects the last two terms over one horizon; measured on the canonical run, both remain below one percent of the transmitted moment (maxima 0.84% and 0.55%, Section IV-D). The reduced moment then splits, using $m\dot{\mathbf{v}}_{\text{com}} = \sum_j \mathbf{f}_j$ from (IV-D.2a), into

$$\mathbf{M}_{s \rightarrow r} \approx \dot{\mathbf{L}}_{\text{robot}} = \underbrace{\dot{\mathbf{L}}_{\text{com}}}_{\text{spin rate}} + \underbrace{\mathbf{r}_{\text{com}} \times \sum_{j=1}^{n_c} \mathbf{f}_j}_{\text{orbital term}}, \quad (\text{V-B.2c})$$

the centroidal *spin rate* and an *orbital term* that grows with the standoff $\|\mathbf{r}_{\text{com}}\|$. The contact-wrench moment (V-B.2) is the reduced robot-induced moment that the momentum-exchange actuator is commanded to absorb. Holding the structure requires the wheels to counter $\boldsymbol{\tau}_{r \rightarrow s}$, so their stored momentum changes at $\boldsymbol{\tau}_w = \boldsymbol{\tau}_{r \rightarrow s} = -\dot{\mathbf{L}}_{\text{robot}}$ (V-A.5), up to the transport, moving-origin, structure-momentum and external-torque residuals stated above. This is the quantity constrained online by the predictive controller of Section VI. The control inputs $\boldsymbol{\tau}_r$ and contact wrench $\boldsymbol{\lambda}$ drive $\mathbf{M}_{s \rightarrow r}$ through the KKT system (IV-C.4).

Combined with the absorption requirement (V-A.5), the two saturation conditions define the time-varying feasible control set:

$$\mathcal{U}_{\text{feasible}}(t) = \left\{ (\boldsymbol{\tau}_r, \boldsymbol{\lambda}) \left| \begin{array}{l} \|\mathbf{h}_w(t)\|_{\infty} \leq h_{\text{max}} \\ \|\boldsymbol{\tau}_{r \rightarrow s}\|_{\infty} \leq \tau_{w, \text{max}} \end{array} \right. \right\}. \quad (\text{V-B.3})$$

What closes as $\|\mathbf{h}_w\| \rightarrow h_{\text{max}}$ is not $\mathcal{U}_{\text{feasible}}(t)$ itself but the subset of it that remains *viable*: inputs that keep injecting momentum along the saturating axis leave no admissible continuation once the store is full. That margin over the remaining step is not represented by an instantaneous torque-only controller with no storage-viability model, which is what motivates the predictive approach of Section VI.

C. MOMENTUM STORAGE ENVELOPE AND SATURATION CONSTRAINTS

Reduction to an equivalent per-axis orthogonal model

The set of physically admissible stored momenta depends on the actuation architecture. For a redundant pyramid RWA ($n_w = 4$, skewed axes), $\mathbf{h}_w = \mathbf{B}_w \dot{\mathbf{q}}_w$ spans a zonotope with oblique facets. For a CMG cluster, the admissible region is directional and contains internal singularities [23]. In both cases, a conservative *per-axis orthogonal equivalent* is obtained by projecting the architecture's net angular momentum onto the three body axes of $\mathcal{R}_s$, yielding a symmetric axis-aligned box:

$$\|\mathbf{h}_w\|_{\infty} \leq h_{\text{max}}, \quad (\text{V-C.1})$$

where h_{max} is the spin-momentum capacity allocated to this robot rather than a value fitted to a particular actuator. Likewise, $\tau_{w, \text{max}}$ below is interpreted as a robot-induced reaction allocation at the planning layer. In the present simulation it is numerically equal to the physical per-axis wheel command clamp, but no feedback reserve is subtracted; therefore satisfaction of the planner bound does not imply satisfaction of the total pre-clipping AOCS command. This conservative inscribed box is exact for a three-axis orthogonal RWA ($n_w = 3$, $\mathbf{B}_w = \text{diag}(I_{w,1}, I_{w,2}, I_{w,3})$). For a CMG cluster it can serve as a planner-side allocation only once the box is verified to lie inside a jointly feasible, singularity-free torque-momentum region (Section VIII); the projection itself is what decouples the planner from the steering-law details.

The per-axis form has a second property that motivates it beyond conservatism: virtual per-robot infinity-norm allocations compose additively, so a host's capacity can be partitioned into per-robot shares. The accounting layer required to enforce such virtual shares is not implemented here (Section VIII).

Conservation-based bilinear bound (theoretical basis)

In the absence of external torques, the total angular momentum is conserved (Section V-A). Written with the robot momentum reduced at the structure CoM through (V-B.1) and the AOCS momentum identified with the stored wheel momentum $\mathbf{h}_w$, the exact three-subsystem balance of (V-A.1) reads, in its $\mathcal{R}_s$ coordinates,

$$\mathbf{h}_w + \mathbf{L}_{\text{robot}} + \mathbf{L}_s = \mathbf{c}^S(t), \quad \mathbf{L}_{\text{robot}} = \mathbf{L}_{\text{com}} + \mathbf{r}_{\text{com}} \times m\mathbf{v}_{\text{com}}, \quad (\text{V-C.2})$$

where $\mathbf{L}_s = \mathbf{I}_s \boldsymbol{\omega}_s$ is the structure's own angular momentum and $\mathbf{c}^S(t)$ the total momentum of (V-A.1): an inertial constant whose $\mathcal{R}_s$ coordinates vary only through the frame transport of (V-A.2).

The planner does not carry $\mathbf{L}_s$. Refreshing a reduced offset once per control cycle gives

$$\mathbf{c}_{\text{simple}}(k) = \mathbf{h}_w(k) + \mathbf{L}_{\text{robot}}(k) = \mathbf{c}^S(k) - \mathbf{L}_s(k), \quad (\text{V-C.3})$$

computed from current measurements and passed as a *parameter* to the NLP — not as a state or decision variable. The frozen offset is not a globally conserved constant but a

local reduced-momentum offset, exact at the refresh instant by construction, whatever $\mathbf{L}_s(k)$ happens to be. A small attitude excursion does not by itself imply a small $\mathbf{L}_s$. Over the horizon the storage prediction carries an error $\mathbf{h}_w(t_i) - \hat{\mathbf{h}}_w(t_i) = \mathbf{L}_s(t_k) - \mathbf{L}_s(t_i)$, plus the frame-transport and external-torque residuals. What the reduction needs is thus that the structure's own angular momentum change *little over one prediction horizon*, which the bounded reaction plan and the separately executed attitude controller jointly support: by bounding the moment transmitted to the structure it bounds the rate at which $\mathbf{L}_s$ can move. The reduction is in that sense self-consistent with the behaviour it presupposes, and the residual it leaves is a measurable quantity rather than an assumed one.

Solving (V-C.3) for the wheel momentum expresses its horizon prediction entirely in robot-centroidal quantities and the frozen offset,

$$\hat{\mathbf{h}}_{w,i|k} = \mathbf{c}_{\text{simple}}(k) - \mathbf{L}_{\text{com},i} - \mathbf{r}_{\text{com},i} \times m\mathbf{v}_{\text{com},i}, \quad (\text{V-C.4})$$

so substituting into (V-C.1) yields a *bilinear* path constraint on the centroidal state $(\mathbf{r}_{\text{com}}, \mathbf{v}_{\text{com}}, \mathbf{L}_{\text{com}})$. The cross-product term $\mathbf{r}_{\text{com}} \times m\mathbf{v}_{\text{com}}$ is bilinear in the state, so (V-C.5a) is a vector ∞ -norm bound on a bilinear function of it: the admissible region it carves in the state space is *not* convex. Neither linearisation nor convex relaxation is applied; the constraint is passed as written to the interior-point solver IPOPT [24], with the local-optimality guarantees that a nonlinear program affords.

In practice the reduced offset varies from cycle to cycle, because the omitted structure-momentum, frame-transport and external-torque residuals vary. Refreshing it at each NMPC cycle through (V-C.3) keeps the constraint consistent with the actual state at every solve.

Rate constraint

Combined with the reaction-torque bound from (V-B.3), the two hard path constraints read as follows over their respective index sets. The storage bound is a state constraint and holds at all $N + 1$ knots $i = 0, \dots, N$, while the rate bound is a function of the control $\mathbf{u}_i$ and is therefore defined only on $i \in [0, N - 1]$:

$$\|\mathbf{c}_{\text{simple}}(k) - \mathbf{L}_{\text{com},i} - \mathbf{r}_{\text{com},i} \times m\mathbf{v}_{\text{com},i}\|_{\infty} \leq h_{\text{max}}, \quad (\text{V-C.5a})$$

$$\|\boldsymbol{\tau}_{r \rightarrow s,i}\|_{\infty} \leq \tau_{w,\text{max}}, \quad (\text{V-C.5b})$$

where $\boldsymbol{\tau}_{r \rightarrow s,i}$ is the moment the robot transmits to the structure about its centre of mass, and is the dominant feedforward quantity the attitude actuator must counter to hold the structure. For deployment with a guaranteed feedback reserve $\tau_{\text{fb},\text{res}} \geq 0$, the planner-side box should be tightened componentwise to

$$|\tau_{r \rightarrow s,i,a}| \leq \tau_{w,\text{max},a}^{\text{phys}} - \tau_{\text{fb},\text{res},a}, \quad a \in \{x, y, z\}, \quad (\text{V-C.6})$$

provided the reserved feedback bound is itself justified. The reported canonical experiment corresponds to zero reserve, so

(V-C.6) is stated as the hardware-facing form of the interface rather than as a property validated by the present run. Its sign follows from the wrench convention: throughout, $\boldsymbol{\lambda}_j = [\mathbf{f}_j, \boldsymbol{\tau}_j]$ is the wrench exerted *by the structure on the robot*, which is the convention the centroidal dynamics of (IV-D.2b) uses. The moment of those wrenches about the structure CoM is the contact-wrench moment $\mathbf{M}_{s \rightarrow r,i}$ of (V-B.2), with the lever $\mathbf{r}_{C_j,i}$ measured from the *structure CoM* (the origin of $\mathcal{R}_s$), and the reaction transmitted to the structure is its negative, $\boldsymbol{\tau}_{r \rightarrow s,i} = -\mathbf{M}_{s \rightarrow r,i}$. Because the bound is symmetric, constraining $\boldsymbol{\tau}_{r \rightarrow s,i}$ and constraining $\mathbf{M}_{s \rightarrow r,i}$ are the same constraint, and it is the latter that the implementation evaluates. Both (V-C.5a) and (V-C.5b) are hard *planner* path constraints, satisfied by construction at every applicable knot of a feasible NMPC solution. They are not a hard constraint on the total executed AOCS command. Their objects differ: the rate bound acts on the planned reaction itself, while the storage bound acts on the *reduced prediction* $\hat{\mathbf{h}}_w$ built from $\mathbf{c}_{\text{simple}}(k)$ — physical satisfaction is verified empirically and re-established at each refresh. In the simulations of Section VII the rate constraint is the *binding* limit, reached at feasible single-support knots without violation, whereas the reduced predicted-storage constraint remains inactive. The measured physical wheel momentum peaks at 82% of the allocated box on the binding axis; the orbital scaling governing this margin is analysed in Section VIII.

Frugal parametric coupling with the attitude-control subsystem

A key architectural consequence of (V-C.3) is that the two optimisation layers receive *different* frozen momentum parameters: Stage 1 the reduced offset $\mathbf{c}_{\text{simple}}(k) \in \mathbb{R}^3$, computed from measurements and frozen for each solve, the whole-body QP the measured wheel momentum $\mathbf{h}_w(k)$ directly (Section VI-C). Neither is a state or decision variable. The centroidal NMPC therefore requires no model of the attitude-control subsystem — allocation law, steering logic, or unloading strategy — and the attitude controller receives no reference from the locomotion layer; the only channels are the one-way $\mathbf{h}_w$ measurement and the contact wrenches supplied to the attitude feedforward (Section VI-D). The coupling is a *frozen per-cycle parametric exchange*: the wheel dynamics are never co-integrated into either optimisation, which is what distinguishes the formulation from approaches that carry wheel momentum in the NMPC state.

VI. TWO-STAGE HIERARCHICAL CONTROL ARCHITECTURE

A. MOTIVATION AND ARCHITECTURE OVERVIEW

As shown in Section IV-D, solving the full KKT system (IV-C.4) over a planning horizon is incompatible with the present 10 Hz implementation and planning budget. Enforcing the horizon-wide constraints (V-C.5a) and (V-C.5b) therefore requires a reduced-order predictive layer. We adopt the two-stage hierarchical decomposition [21], [25] (Fig. 3):

- **Stage 1 — Centroidal NMPC (10 Hz):** reduced-order centroidal model, $N = 8$, $\Delta t = 0.1$ s; generates a constrained centroidal state and contact-wrench plan; solved via CasADi [26]/IPOPT [24] (Section VII-G).
- **Stage 2 — Whole-Body QP (100 Hz):** full multibody dynamics; realises the plan through the CoM linear-momentum, 6-D torso-pose, swing end-effector and posture tasks (Section VI-C).

a: Hybrid SS/DS architecture.

The centroidal NMPC generates locomotion references during the *single-support (SS) phases* and remains applied during the *terminal double-support hold* after the final weld, where the two-contact set is frozen and the solver regulates the centroidal state in place rather than generating a new stride (Section VII-A). During the *inter-step double-support (DS) transitions* both arms are anchored and the torso is held near the docked configuration by the whole-body QP; there — and only there — the NMPC is bypassed, and it is re-engaged at the start of the subsequent SS phase with a fresh warm start. The reaction wheels are *not* idle during DS: the attitude controller (Section VI-D) commands them at every control cycle, because the free-floating structure must be attitude-regulated continuously.

B. STAGE 1 – CENTROIDAL NMPC

State and control vectors

The centroidal state introduced in Section IV-D provides the NMPC planning variable. Using the centroidal dynamics (IV-D.2a)–(IV-D.2b), the state and control vectors are:

$$\mathbf{x} = \begin{bmatrix} \mathbf{r}_{\text{com}} \\ \mathbf{v}_{\text{com}} \\ \mathbf{L}_{\text{com}} \end{bmatrix} \in \mathbb{R}^9, \quad \mathbf{u} = \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} \in \mathbb{R}^{12}, \quad (\text{VI-B.1})$$

where each contact wrench $\lambda_j = [\mathbf{f}_j^\top, \boldsymbol{\tau}_j^\top]^\top \in \mathbb{R}^6$ is the Lagrange multiplier of (IV-B.1) for contact j . The control vector is sized to the maximum contact count; in single support the released arm's wrench is held at zero, so the planned reaction is the moment of the stance contact alone. The wheel momentum $\mathbf{h}_w$ enters only through the frozen parameter $\mathbf{c}_{\text{simple}}(k)$ (V-C.3) (Section V-C).

Prediction model

The centroidal dynamics (IV-D.2a)–(IV-D.2b) are discretised with RK4 at $\Delta t = 0.1$ s, giving the discrete transition $\mathbf{x}_{i+1} = f_{\text{RK4}}(\mathbf{x}_i, \mathbf{u}_i)$.

NMPC formulation

At each 10 Hz cycle k the NMPC solves:

$$\begin{aligned} \min_{\mathbf{x}_{0:N}, \mathbf{u}_{0:N-1}} \sum_{i=0}^{N-1} & \left[\|\mathbf{r}_{\text{com},i} - \hat{\mathbf{r}}_{\text{com},i}\|_{\mathbf{W}_r}^2 + \|\mathbf{v}_{\text{com},i} - \hat{\mathbf{v}}_{\text{com},i}\|_{\mathbf{W}_v}^2 \right. \\ & \left. + w_L \|\mathbf{L}_{\text{com},i} - \hat{\mathbf{L}}_{\text{com},i}\|^2 + \|\mathbf{u}_i\|_{\mathbf{W}_u}^2 \right] \\ & + \|\mathbf{r}_{\text{com},N} - \hat{\mathbf{r}}_{\text{com},N}\|_{\mathbf{Q}_{f,r}}^2 \\ & + \|\mathbf{v}_{\text{com},N} - \hat{\mathbf{v}}_{\text{com},N}\|_{\mathbf{Q}_{f,v}}^2 \\ & + \mathbf{Q}_{f,L} \|\mathbf{L}_{\text{com},N} - \hat{\mathbf{L}}_{\text{com},N}\|^2, \end{aligned} \quad (\text{VI-B.2})$$

subject to, for every contact j :

$$\mathbf{x}_{i+1} = f_{\text{RK4}}(\mathbf{x}_i, \mathbf{u}_i), \quad \mathbf{x}_0 = \mathbf{x}(t_k), \quad (\text{VI-B.3a})$$

$$\|\mathbf{c}_{\text{simple}}(k) - \mathbf{L}_{\text{com},i} - \mathbf{r}_{\text{com},i} \times m\mathbf{v}_{\text{com},i}\|_{\infty} \leq h_{\text{max}}, \quad (\text{VI-B.3b})$$

$$\|\boldsymbol{\tau}_{r \rightarrow s,i}\|_{\infty} \leq \tau_{w,\text{max}}, \quad (\text{VI-B.3c})$$

$$\|\mathbf{f}_{i,j}\|_2 \leq f_{\text{max}}, \quad \|\boldsymbol{\tau}_{i,j}\|_2 \leq \tau_{\text{max}}, \quad (\text{VI-B.3d})$$

$$\|\mathbf{L}_{\text{com},i}\|_{\infty} \leq L_{\text{max}}, \quad \|m\mathbf{v}_{\text{com},i}\|_2 \leq p_{\text{max}}. \quad (\text{VI-B.3e})$$

The numerical cost weights of the frozen configuration are $\mathbf{W}_r = 100\mathbf{I}_3$, $\mathbf{W}_v = 10\mathbf{I}_3$, $\mathbf{W}_u = \text{diag}(0.01\mathbf{I}_3, 0.001\mathbf{I}_3, 0.01\mathbf{I}_3, 0.001\mathbf{I}_3)$, $\mathbf{Q}_{f,r} = 1000\mathbf{I}_3$ and $\mathbf{Q}_{f,v} = 100\mathbf{I}_3$. The stage cost tracks $\mathbf{L}_{\text{com}}$ with $w_L = 1.0$ and the terminal cost uses $\mathbf{Q}_{f,L} = 10.0$. The implementation additionally enforces $L_{\text{max}} = 10$ N·m·s and $p_{\text{max}} = 50$ kg·m/s in (VI-B.3e); these auxiliary bounds were present in the frozen controller and are stated here explicitly for reproducibility. The angular momentum reference $\hat{\mathbf{L}}_{\text{com}} = \mathbf{I}_{\text{torso}} \hat{\boldsymbol{\omega}}_{\text{torso}}(t)$ is provided by the TorsoPlanner at each cycle. The CoM references $\hat{\mathbf{r}}_{\text{com}}$, $\hat{\mathbf{v}}_{\text{com}}$ are computed from the gait scheduler and IK pipeline. Constraints (VI-B.3b)–(VI-B.3c) are the hard path constraints of Section V-C. The storage constraint (VI-B.3b) is enforced at all $N + 1$ state knots, including the terminal state. The rate constraint (VI-B.3c), being a function of $\mathbf{u}_i$, is enforced over $i = 0, \dots, N - 1$. The first optimal control $\mathbf{u}_0^*$ provides the wrench reference to Stage 2, while the optimal state trajectory $\mathbf{x}_{0:N}^*$ provides the CoM state reference and seeds the warm start at $k + 1$.

Scope of the guarantee. The per-axis bounds are guaranteed *by construction* only for the objects and locations that occur in the nonlinear programme: the reduced storage prediction at the $N + 1$ state knots and the planned robot-induced reaction at the N control knots of each feasible NMPC solve. This statement does not imply recursive feasibility at the next receding-horizon update, continuous-time satisfaction between knots, satisfaction during inter-step double support (where the predictive layer is bypassed), or satisfaction of the *total* pre-clipping AOCs command, which also contains feedback. The terminal double-support hold is an implementation mode in which the NMPC continues to solve on a frozen contact set. All quantities outside the formal planner guarantee are reported as empirical measurements in Section VII; no stronger guarantee is claimed.

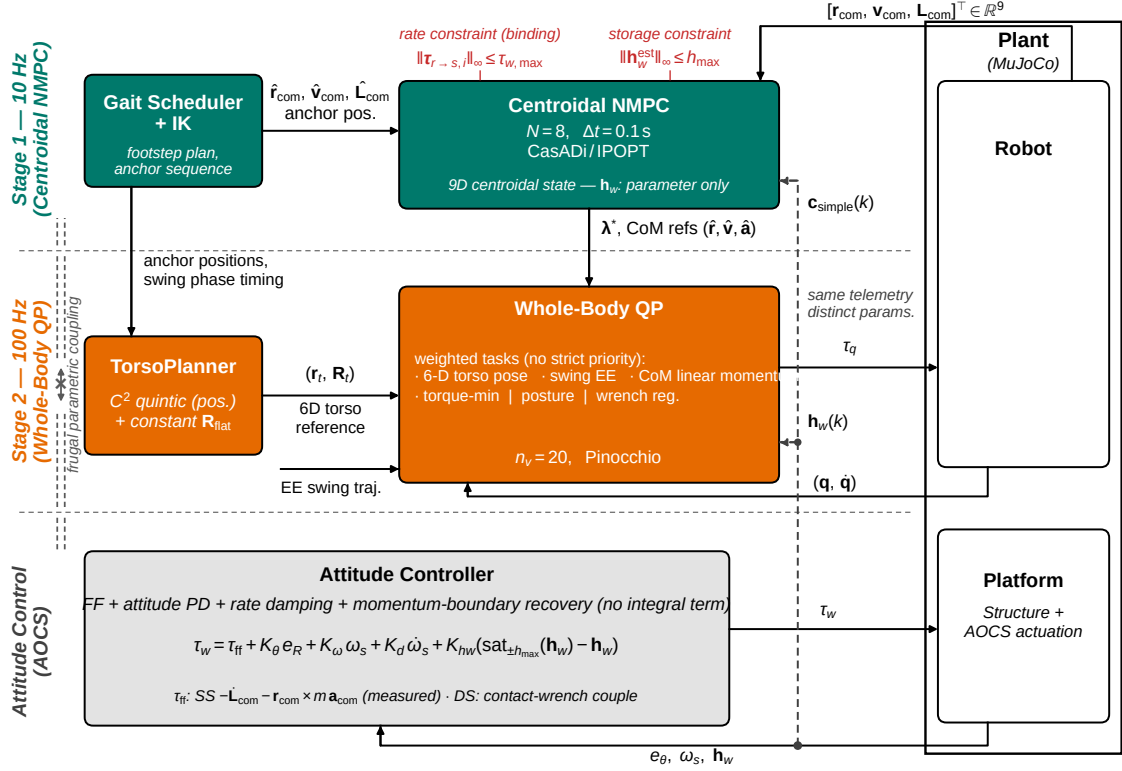


FIGURE 3. Two-stage hierarchical control architecture. **Stage 1** (teal, 10 Hz): centroidal NMPC with 9D state $[\mathbf{r}_{\text{com}}, \mathbf{v}_{\text{com}}, \mathbf{L}_{\text{com}}]^T$; two hard planner path constraints bound the allocated robot-induced reaction and the reduced storage prediction. **Stage 2** (orange, 100 Hz): whole-body QP realising the plan through a weighted task stack with no strict priority projection ($n_v = 20$, Pinocchio); it tracks the CoM linear-momentum dynamics, the 6-D torso pose, the swing end-effector and the NMPC wrench reference. **AOCS** (grey): independent feedforward-PD attitude controller. The stored momentum $\mathbf{h}_w$ enters the locomotion layers only as the frozen per-cycle parameters $\mathbf{c}_{\text{simple}}(k)$ (NMPC) and $\mathbf{h}_w(k)$ (QP), shown dashed.

C. STAGE 2 – WHOLE-BODY QP

Decision variables and dynamics

Stage 2 computes joint commands from the robot block of (IV-C.4):

$$\tilde{\mathbf{H}}_t \ddot{\mathbf{q}}_t + \tilde{\mathbf{H}}_{tq} \ddot{\mathbf{q}} + \tilde{\mathbf{c}}_t = (\tilde{\mathbf{J}}^t)^T \boldsymbol{\lambda}, \quad (\text{VI-D.1})$$

$$\mathbf{H}_q \ddot{\mathbf{q}} + \tilde{\mathbf{H}}_{tq}^T \ddot{\mathbf{q}}_t + \tilde{\mathbf{c}}_q = \boldsymbol{\tau}_q + (\tilde{\mathbf{J}}^q)^T \boldsymbol{\lambda}, \quad (\text{VI-D.2})$$

the torso and joint rows of the robot block of (IV-C.4), with $\tilde{\mathbf{H}}_t$, $\tilde{\mathbf{H}}_{tq}$, $\mathbf{H}_q$ the inertia blocks of (IV-C.2), $\tilde{\mathbf{J}}^t$, $\tilde{\mathbf{J}}^q$ the contact-Jacobian blocks of (IV-A.2) for the active contacts, and $\tilde{\mathbf{c}}_t$, $\tilde{\mathbf{c}}_q$ the partitions of the velocity term $\mathbf{c}_r$. The QP decision vector is:

$$\mathbf{z} = [\ddot{\mathbf{q}}_t^T, \ddot{\mathbf{q}}^T, \boldsymbol{\lambda}^T, \boldsymbol{\tau}_q^T]^T \in \mathbb{R}^{6+n_q+6n_c+n_q}, \quad (\text{VI-D.3})$$

whose first two blocks form the robot acceleration $\ddot{\mathbf{q}}_r = [\ddot{\mathbf{q}}_t^T, \ddot{\mathbf{q}}^T]^T$ on which the tasks below are written. The physical decision vector has dimension $6 + 14 + 12 + 14 = 46$ for $n_q = 14$ and $n_c = 2$ (the wrench block is sized to the maximum contact count). Joint torques $\boldsymbol{\tau}_q$ are direct decision variables, allowing torque bounds to be enforced as simple box constraints. The implementation augments $\mathbf{z}$ with non-negative slacks $\mathbf{s}_{\text{up}}, \mathbf{s}_{\text{lo}} \in \mathbb{R}^3$ that soften the one-step spin-only safeguard band (below), for 52 solved variables; the slacks are pinned to zero whenever that row is inactive, and

their measured activity on the canonical run is reported with the band below.

Three-task weighted stack

Stage 2 is a single weighted quadratic program: tasks are summed with weights, not projected into null spaces. Three tracking tasks are driven from *direct* sources — each reference is generated independently of the state it commands, so no task closes an algebraic loop:

- **CoM linear-momentum / acceleration task** (α_{mom}). Using the centroidal momentum matrix $\mathbf{A}_G(\mathbf{q}) \in \mathbb{R}^{6 \times n_v}$, which maps the generalised velocity to the centroidal momentum $\mathbf{h}_G = \mathbf{A}_G \dot{\mathbf{q}}_r = [m \mathbf{v}_{\text{com}}^T, \mathbf{L}_{\text{com}}^T]^T$, the whole-body centre of mass is regulated at the acceleration level:

$$\mathbf{A}_{G,v} \ddot{\mathbf{q}}_r + \dot{\mathbf{A}}_{G,v} \dot{\mathbf{q}}_r = m \mathbf{a}_{\text{com}}^{\text{des}}, \quad (\text{VI-D.4})$$

Let $\mathbf{e}_r = \hat{\mathbf{r}}_{\text{com}} - \mathbf{r}_{\text{com}}$ and $\mathbf{e}_v = \hat{\mathbf{v}}_{\text{com}} - \mathbf{v}_{\text{com}}$. The desired acceleration is

$$\mathbf{a}_{\text{com}}^{\text{des}} = \hat{\mathbf{a}}_{\text{com}} + \mathbf{K}_p^{\text{com}} \mathbf{e}_r + \mathbf{K}_d^{\text{com}} \mathbf{e}_v,$$

with axis-decoupled block-diagonal gains

$$\mathbf{K}_\ell^{\text{com}} = \text{blkdiag}(k_{\ell,x}^{\text{com}}, k_{\ell,y}^{\text{com}}, k_{\ell,z}^{\text{com}}), \quad \ell \in \{p, d\}.$$

Here $\hat{\mathbf{r}}_{\text{com}}$, $\hat{\mathbf{v}}_{\text{com}}$ and $\hat{\mathbf{a}}_{\text{com}}$ are the NMPC optimal plan (Section VI-B), and $\mathbf{A}_{G,v}$ denotes the linear (translational-momentum) rows of $\mathbf{A}_G$. The canonical tuning is isotropic, $\mathbf{K}_p^{\text{com}} = \mathbf{K}_d^{\text{com}} = 3\mathbf{I}_3$; each Cartesian error component therefore acts only on its corresponding desired-acceleration component. Coupling induced by the multibody dynamics, contacts, and the weighted QP remains represented by the assembled optimisation problem.

- **6-D torso-pose task** (α_{torso}). Because the arm joints do not move the torso frame, the task Jacobian has non-zero entries only in the base block:

$$\mathbf{J}_{\text{torso}} = [\tilde{\mathbf{J}}_{\text{torso}}^t \quad \mathbf{0}_{6 \times 7} \quad \mathbf{0}_{6 \times 7}] \in \mathbb{R}^{6 \times n_v},$$

with $\tilde{\mathbf{J}}_{\text{torso}}^t$ the base block of (IV-A.2) evaluated at the torso frame. The task tracks the position (A.1) (Appendix A) and constant orientation (A.2) references directly:

$$\mathbf{J}_{\text{torso}} \ddot{\mathbf{q}}_r + \dot{\mathbf{J}}_{\text{torso}} \dot{\mathbf{q}}_r = \ddot{\mathbf{x}}_{\text{torso}}^{\text{des}}, \quad (\text{VI-D.5})$$

with $\ddot{\mathbf{x}}_{\text{torso}}^{\text{des}} = \ddot{\mathbf{x}}^{\text{ref}} + \mathbf{K}_d^{\text{torso}}(\dot{\mathbf{x}}^{\text{ref}} - \dot{\mathbf{x}}) + \mathbf{K}_p^{\text{torso}} \mathbf{e}_6(\mathbf{x}^{\text{ref}}, \mathbf{x})$, where $\mathbf{e}_6$ stacks the position difference and the body-frame orientation error $\log(\mathbf{R}^T \mathbf{R}^{\text{ref}})$ ($\mathbf{K}_p^{\text{torso}} = 3$, $\mathbf{K}_d^{\text{torso}} = 2.5$).

- **6-D swing end-effector task** (α_{ee}). This is the task that performs the step. For the swinging chain j , (IV-A.2) gives the gripper twist as $\tilde{\mathbf{J}}_t^j \dot{\mathbf{q}}_t + \tilde{\mathbf{J}}_q^j \dot{\mathbf{q}}^j$; $\mathbf{J}_{ee} \in \mathbb{R}^{6 \times n_v}$ assembles those blocks against the full $\dot{\mathbf{q}}_r$, the zero block occupying the stance chain's columns. With the column order of $\dot{\mathbf{q}}_r = [\dot{\mathbf{q}}_t^T \dot{\mathbf{q}}^{A^T} \dot{\mathbf{q}}^{B^T}]^T$ and swing chain j :

$$\mathbf{J}_{ee} = [\tilde{\mathbf{J}}_t^j \quad \mathbf{0}_{6 \times 7} \quad \tilde{\mathbf{J}}_q^j] \quad (j = B \text{ shown}),$$

$$\mathbf{J}_{ee} \dot{\mathbf{q}}_r = \tilde{\mathbf{J}}_t^j \dot{\mathbf{q}}_t + \tilde{\mathbf{J}}_q^j \dot{\mathbf{q}}^j.$$

During single support it tracks the swing reference (A.3), so that both the position and the orientation gates of the docking condition are driven, not only the approach distance:

$$\mathbf{J}_{ee} \ddot{\mathbf{q}}_r + \dot{\mathbf{J}}_{ee} \dot{\mathbf{q}}_r = \ddot{\mathbf{x}}_{ee}^{\text{des}}, \quad (\text{VI-D.6})$$

with $\ddot{\mathbf{x}}_{ee}^{\text{des}}$ built by the same feedforward-plus-pose-error-PD structure as the torso task, on the swing reference. The task is carried only through single support: in double support both grippers are seated by (IV-B.1).

The swing task sets what the step must achieve; the torso-pose task holds the torso against the reaction that step provokes; the CoM linear-momentum task governs, at the *rate* level, the residual CoM motion the pose task leaves free. The consequence — the CoM *position* lags its reference during dynamic swings — is quantified in Section VII and motivates the path-following direction of Section VIII.

The complete weighted objective is:

$$\begin{aligned} \min_{\mathbf{z}} \quad & \frac{1}{2} \left[\alpha_{\text{mom}} \|\mathbf{A}_{G,v} \ddot{\mathbf{q}}_r + \dot{\mathbf{A}}_{G,v} \dot{\mathbf{q}}_r - m \mathbf{a}_{\text{com}}^{\text{des}}\|^2 \right. \\ & + \alpha_{\text{torso}} \|\mathbf{J}_{\text{torso}} \ddot{\mathbf{q}}_r + \dot{\mathbf{J}}_{\text{torso}} \dot{\mathbf{q}}_r - \ddot{\mathbf{x}}_{\text{torso}}^{\text{des}}\|^2 \\ & + \alpha_{ee} \|\mathbf{J}_{ee} \ddot{\mathbf{q}}_r + \dot{\mathbf{J}}_{ee} \dot{\mathbf{q}}_r - \ddot{\mathbf{x}}_{ee}^{\text{des}}\|^2 + \alpha_{\text{posture}} \|\ddot{\mathbf{q}} - \ddot{\mathbf{q}}_{\text{ref}}\|^2 \\ & + \alpha_{\text{wrench}} \|\boldsymbol{\lambda} - \boldsymbol{\lambda}_{\text{ref}}\|^2 + \alpha_{\tau} \|\boldsymbol{\tau}_q\|^2 + \alpha_{\text{reg}} \|\ddot{\mathbf{q}}\|^2 \\ & \left. + \alpha_{\text{int}} \|\mathbf{P}_{\text{int}} \boldsymbol{\lambda}\|^2 + w_h (\|\mathbf{s}_{\text{up}}\|^2 + \|\mathbf{s}_{\text{lo}}\|^2) \right], \quad (\text{VI-D.7}) \end{aligned}$$

where $\ddot{\mathbf{q}}_{\text{ref}} = \mathbf{K}_p^{\text{post}}(\mathbf{q}_0 - \mathbf{q}) - \mathbf{K}_d^{\text{post}} \dot{\mathbf{q}}$ is a posture PD resolving the arm redundancy, and $\alpha_{\tau} \|\boldsymbol{\tau}_q\|^2$ a torque-minimisation term ($\alpha_{\tau} = 5$ against $\alpha_{\text{reg}} = \alpha_{\text{wrench}} = 1$). The α_{int} term is the internal-stress task defined with the momentum maps below. The final term penalises the slacks $\mathbf{s}_{\text{up}}, \mathbf{s}_{\text{lo}}$ of the spin-only safeguard band: their bounds open only when the one-step row of (VI-D.8) would otherwise leave its $\pm h_{\text{qp}}$ band, and they are pinned to zero otherwise. On the run reported here the lower slack opens on 389 of the 8458 solves (4.6%), with a maximum of 1.11 N·m·s, and the upper slack never activates: the band is a soft early-warning row, relaxed exactly when the measured momentum exceeds it, while the allocated $\pm h_{\text{max}}$ box itself is never breached (Section VII). The assembled problem solved at each cycle is

$$\begin{aligned} \min_{\mathbf{z}} \quad & J(\mathbf{z}) \text{ of (VI-D.7)} \\ \text{s.t.} \quad & \mathbf{H}_r \ddot{\mathbf{q}}_r + \mathbf{c}_r = \mathbf{S}^T \boldsymbol{\tau}_q + \mathbf{J}_c^T \boldsymbol{\lambda}, \\ & \mathbf{J}_c \ddot{\mathbf{q}}_r + \dot{\mathbf{J}}_c \dot{\mathbf{q}}_r = \mathbf{0}, \quad \boldsymbol{\lambda}_j = \mathbf{0} \quad (j \text{ inactive}), \\ & \|\mathbf{M}_{\lambda}^O \boldsymbol{\lambda}\|_{\infty} \leq \tau_{w,\text{max}}, \\ & -h_{\text{qp}} - \mathbf{s}_{\text{lo}} \leq \mathbf{h}_w - \Delta t \mathbf{M}_{\lambda}^{\text{com}} \boldsymbol{\lambda} \leq h_{\text{qp}} + \mathbf{s}_{\text{up}}, \\ & \mathbf{s}_{\text{up}}, \mathbf{s}_{\text{lo}} \geq \mathbf{0}, \\ & \boldsymbol{\tau}_{\min} \leq \boldsymbol{\tau}_q \leq \boldsymbol{\tau}_{\max}, \quad \boldsymbol{\lambda}_{\min} \leq \boldsymbol{\lambda} \leq \boldsymbol{\lambda}_{\max}, \\ & \ddot{\mathbf{q}}_{\min} \leq \ddot{\mathbf{q}} \leq \ddot{\mathbf{q}}_{\max}, \end{aligned} \quad (\text{VI-D.8})$$

with $\mathbf{H}_r, \mathbf{c}_r$ the robot inertia and velocity terms of (IV-C.2), $\mathbf{S}$ the joint selection matrix, $\mathbf{J}_c$ the stacked contact Jacobian of the active contacts, and the bounds listed in Table 3. The first inequality is the Stage-2 envelope: it holds the wrench the QP may request inside the same per-axis bound the NMPC plans against, and it is what limits the *realised* reaction torque whatever Stage 1 requested.

Two momentum maps appear, differing only in their reduction point,

$$\begin{aligned} \mathbf{M}_{\lambda}^p &= [\mathbf{S}(\mathbf{r}_{C_A} - \mathbf{p}) \quad \mathbf{I}_3 \quad \mathbf{S}(\mathbf{r}_{C_B} - \mathbf{p}) \quad \mathbf{I}_3] \in \mathbb{R}^{3 \times 12}, \\ \mathbf{M}_{\lambda}^O(\mathbf{p} = \mathbf{0}), \quad \mathbf{M}_{\lambda}^{\text{com}}(\mathbf{p} = \mathbf{r}_{\text{com}}), \end{aligned}$$

with $\mathbf{S}(\cdot)$ the skew map, the columns of an inactive contact set to zero, and $\mathbf{r}_{C_j}$ expressed in $\mathcal{R}_s$; both are recomputed every cycle from the current configuration and act on the same structure-on-robot wrenches. The envelope constraint uses the origin-referenced map, so it bounds the moment about the structure centre of mass exactly. The soft one-step row uses the CoM-referenced map and therefore omits the orbital term $\mathbf{r}_{\text{com}} \times \sum_j \mathbf{f}_j$: it is a *spin-only storage safeguard*, not

a prediction of the full stored momentum. Its band is tighter than the allocated box ($h_{qp} = 3 \text{ N}\cdot\text{m}\cdot\text{s}$ against $h_{\max} = 5$), so it is an early-warning row: the storage result the paper reports rests on the Stage-1 constraint and on the measured wheel momentum, not on this row (Section VIII). This row uses the measured $\mathbf{h}_w(k)$ directly, not $\mathbf{c}_{\text{simple}}(k)$. The reference wrench $\boldsymbol{\lambda}_{\text{ref}}$ is supplied by the NMPC. In double support the two-contact wrench is redundant: only its net effect on the structure is dynamically determined, and the complementary *internal-stress* component stresses the grasp without moving anything. The α_{int} task of (VI-D.7) minimises that component through the wrench-null-space projector $\mathbf{P}_{\text{int}} = \mathbf{I}_{12} - \mathbf{G}^+ \mathbf{G}$, where $\mathbf{G} \in \mathbb{R}^{6 \times 12}$ is the grasp map sending the stacked wrench $\boldsymbol{\lambda} = [\mathbf{f}_A^\top, \boldsymbol{\tau}_A^\top, \mathbf{f}_B^\top, \boldsymbol{\tau}_B^\top]^\top$ to the resultant structure-on-robot wrench about the origin of $\mathcal{R}_s$,

$$\mathbf{G} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0} & \mathbf{I}_3 & \mathbf{0} \\ \mathbf{S}(\mathbf{r}_{C_A}) & \mathbf{I}_3 & \mathbf{S}(\mathbf{r}_{C_B}) & \mathbf{I}_3 \end{bmatrix},$$

with the same skew map $\mathbf{S}(\cdot)$ as above. The task is assembled in double support only ($\alpha_{\text{int}} = 1.0$, Table 3): in single support the active wrench is six-dimensional, $\mathbf{G}$ is square and full rank, and the internal-stress null space is empty. The weights (Table 3) place the torso-pose task above the momentum task ($\alpha_{\text{torso}} = 2000$, $\alpha_{\text{mom}} = 400$), so torso pointing is held tightly while the CoM is governed at the momentum level. The QP is solved with qpOASES through the CasADi `conic` interface, with a working-set recalculation cap of 500, the solver's default termination tolerances, and a Tikhonov regularisation of 10^{-6} applied to the assembled Hessian solver-side. A nominal 5 ms backend CPU limit is configured; the wall-clock timings of Section VII-G include problem assembly and interface overhead and can therefore exceed that figure. The NMPC sets no IPOPT tolerance or iteration-cap override. The reported simulation contains no online infeasibility fallback: all canonical solves return a success status, so fallback behaviour is not validated here. A deployment implementation should retain the last feasible trajectory or enter a constraint-preserving hold if a solve fails, rather than substituting an unconstrained reference. Over the 8458 QP solves of the canonical run the backend reports success on every solve, with a worst equality residual of 1.5×10^{-11} and a worst inequality violation of 1.3×10^{-15} .

D. ATTITUDE CONTROL

The spacecraft attitude is maintained by an on-board controller that is *structurally independent* of the locomotion planner: it receives no reference from the NMPC, and the only information channel toward Stage 1 is the measurement of $\mathbf{h}_w$ used to build $\mathbf{c}_{\text{simple}}$ (Section V-C).

The control law comprises four terms:

$$\begin{aligned} \boldsymbol{\tau}_w = & \boldsymbol{\tau}_{\text{ff}} + K_\theta e_R + K_\omega \boldsymbol{\omega}_s + K_d \dot{\boldsymbol{\omega}}_s \\ & + K_{hw}(\text{sat}_{\pm h_{\max}}(\mathbf{h}_w) - \mathbf{h}_w), \end{aligned} \quad (\text{VI-E.1})$$

where $\boldsymbol{\omega}_s$ is the measured structure rate and e_R the measured attitude error, taken as the skew-symmetric (Lee-McClamroch) map $e_R = \frac{1}{2}(\mathbf{R}_e - \mathbf{R}_e^\top)^\vee$ with $\mathbf{R}_e =$

$\mathbf{R}_s(0)^\top \mathbf{R}_s(t)$, which agrees with the geodesic angle of Section VII-D to better than 10^{-5} over the excursions reported there. The feedback terms enter with a *positive* sign: $\boldsymbol{\tau}_w$ is applied to the wheels and $-\boldsymbol{\tau}_w$ is the reaction on the structure, so a positive error commands the restoring reaction. In detail:

- $\boldsymbol{\tau}_{\text{ff}}$ is a *momentum feedforward* anticipating the robot-induced moment about O . During SS it uses the quasi-inertial estimate $-\dot{\mathbf{L}}_{\text{com}} - \mathbf{r}_{\text{com}} \times m \mathbf{a}_{\text{com}}$, retaining the spin and orbital terms while neglecting the bounded rotational-transport and moving-origin residuals quantified in Section IV-D; both rates are estimated by finite differences of the *measured* centroidal state, and the orbital term is the dominant contribution identified in Section VIII. During DS it is reconstructed from the whole-body QP wrench solution as $-\sum_j (\mathbf{r}_{C_j} \times \mathbf{f}_j + \boldsymbol{\tau}_j)$. In neither phase does it read the NMPC plan: it anticipates the robot-induced disturbance without requiring any model of the locomotion trajectory.
- $K_\theta e_R$ is the attitude proportional feedback ($K_\theta = 1.0 \text{ N}\cdot\text{m}/\text{rad}$).
- $K_\omega \boldsymbol{\omega}_s + K_d \dot{\boldsymbol{\omega}}_s$ is a rate damping term with second-order structure ($K_\omega = 50.0 \text{ N}\cdot\text{m}\cdot\text{s}/\text{rad}$, $K_d = 25.0 \text{ N}\cdot\text{m}\cdot\text{s}^2/\text{rad}$), which suppresses residual oscillation without an integral term.
- $K_{hw}(\text{sat}_{\pm h_{\max}}(\mathbf{h}_w) - \mathbf{h}_w)$ is a wheel-momentum boundary-recovery term [27] ($K_{hw} = 2.0 \text{ s}^{-1}$): the saturation makes the term vanish inside the $\pm h_{\max}$ box, so it acts only to pull an excursion back to the boundary; inside the box the store is not actively unloaded.

The law (VI-E.1) is a **PD controller with feedforward and a wheel-momentum boundary-recovery term**; it contains *no integral term*. It runs at 100 Hz throughout the locomotion cycle, in both SS and DS. During SS the feedforward is estimated from the measured centroidal rates; during DS — inter-step and terminal alike — it is reconstructed from the whole-body QP contact wrenches. The structure is therefore attitude-regulated across the support transitions as well as during locomotion (frugal coupling, Section V-C).

VII. RESULTS AND DISCUSSION

A. SIMULATION SETUP

The multibody dynamics and rigid contacts are simulated in MuJoCo [28]. The robot model [29] (a dual-arm crawler with seven actuated joints per arm, total mass 71.06 kg, $n_q = 14$ actuated joints, generalised-velocity dimension $n_v = 20$), and all kinematic and dynamic quantities used by the controller are evaluated with Pinocchio [30], and the centroidal NMPC is transcribed and solved with CasADi [26]/IPOPT [24]. The canonical configuration is that of Table 3 ($N = 8$ at 10 Hz, whole-body QP at 100 Hz, AOCS law of Section VI-D). All vector quantities below are expressed in the rotating structure body frame $\mathcal{R}_s$, in which the attitude-actuation constraint is also written. The robot crawls outward across six HOTDOCK anchors fixed in $\mathcal{R}_s$. The structure itself is modelled as a single rigid body with prescribed inertial properties, listed in

TABLE 2. Run provenance. All five share the frozen configuration of Table 3 and differ only in the fields shown. “QP rate” is the whole-body envelope constraint on the requested wrench, “QP safeg.” the soft spin-only storage row, “clip” the attitude-controller output saturation, and “hold” the terminal duration after the final weld. The first four rows are used for the ablation of Section VII-C; *rate and clip lifted* is a previously retained comparison configuration.

Run	NMPC		QP		Clip	Hold [s]
	rate	stor.	rate	saferg.		
Canonical	on	on	on	on	on	20
Rate-only	on	off	on	on	on	20
No envelope	off	off	on	on	on	20
Rate/clip lifted	off	off	off	on	off	20
Station keeping	on	on	on	on	on	900

Table 3; its rendered geometry serves contact detection and the anchor layout, and is not the source of those properties.

We evaluate the two-stage controller on a six-step traversal of a planar modular structure at a 1% robot-to-structure mass ratio, the swinging arm alternating between the two arms at each step.

Every number reported below is read from the per-cycle simulation logs of a single frozen configuration, which fixes $\tau_{w,\max} = 2.5 \text{ N}\cdot\text{m}$ in both the controller and the plant together with the whole-body weights of Table 3. Five runs of that configuration are used, listed in Table 2; they differ only in the fields shown there. Each number below is attributed to its run; none is a best case selected across runs.

Two durations must be kept apart: locomotion occupies the first 64.5 s, to the final weld; the logged run continues for a 20 s terminal hold (84.5 s total), which the station-keeping run extends to 900 s. Statements about *the traversal* refer to the first window. Three execution modes must be kept apart, two of them double-support. In *single support* one gripper is welded and the NMPC solves and is applied. In the *inter-step double-support settle* both grippers are welded, the predictive layer is bypassed, and the whole-body QP and attitude controller continue at their own rates. The *terminal hold* is also double-support, but the NMPC continues to solve and its output is applied: the contact set is frozen and the layer regulates the centroidal state in place rather than generating a stride — hence the 201 terminal ticks count as NMPC-active alongside the 508 single-support ones.

B. DOCKING PERFORMANCE

Table 4 summarises the six docking transitions. All complete within the HOTDOCK-inspired capture gate [19]: the end-effector position error at capture is at most 4.99 mm and the orientation error at most 0.14° . These are *at-weld* values: because capture also gates ξ_{cap} (Fig. 2), the closest approach during a swing (Fig. 4) is not in general the at-weld distance.

C. ATTITUDE-ACTUATION ENVELOPE

Fig. 5 addresses the central claim. The reaction torque planned by the NMPC, $\tau_{r \rightarrow s,i} = -[\sum_j (r_{C_j} \times f_j + \tau_j)]_i$ (V-B.2), is held within the hard per-axis bound $\tau_{w,\max} = 2.5 \text{ N}\cdot\text{m}$

TABLE 3. Canonical-run parameters. The structure inertia is diagonal, in $\text{kg}\cdot\text{m}^2$.

Symbol	Description	Value
<i>Centroidal NMPC</i>		
N	Prediction horizon	8
dt_{nmmpc}	NMPC step (10 Hz)	0.1 s
$\tau_{w,\max}$	Per-axis reaction-torque bound	2.5 N·m
$h_{\max}$	Per-axis stored-momentum bound	$\pm 5.0 \text{ N}\cdot\text{m}\cdot\text{s}$
$f_{\max}$	Contact-force bound (per contact)	300 N
$\tau_{\max}$	Contact-moment bound (per contact)	8 N·m
$\mathbf{W}_r$	CoM-position stage weight	100 $\mathbf{I}_3$
$\mathbf{W}_v$	CoM-velocity stage weight	10 $\mathbf{I}_3$
$\mathbf{W}_u$	Wrench stage weight (force/moment blocks, repeated per contact)	$0.01\mathbf{I}_3/0.001\mathbf{I}_3$
$\mathbf{Q}_{f,r}$	CoM-position terminal weight	1000 $\mathbf{I}_3$
$\mathbf{Q}_{f,v}$	CoM-velocity terminal weight	100 $\mathbf{I}_3$
$w_L, \mathbf{Q}_{f,L}$	Angular-momentum weights	1, 10
$L_{\max}$	Auxiliary ℓ_∞ centroidal-angular-momentum bound	10 N·m·s
$p_{\max}$	Auxiliary CoM linear-momentum bound	50 kg·m/s
<i>Whole-body QP</i>		
dt_{qp}	QP step (100 Hz)	0.01 s
α_{torso}	Torso-pose weight	2000
α_{mom}	CoM linear-momentum weight	400
$\mathbf{K}_{\text{com}}^p$	CoM proportional gain matrix	3.0 $\mathbf{I}_3$
$\mathbf{K}_{\text{com}}^d$	CoM derivative gain matrix	3.0 $\mathbf{I}_3$
$\mathbf{K}_{\text{torso}}^p$	Torso-pose proportional gain	3.0
$\mathbf{K}_{\text{torso}}^d$	Torso-pose derivative gain	2.5
α_{ee}	Swing-EE weight	1000
α_{posture}	Posture weight	20
α_{wrench}	Wrench-regularisation weight	1.0
α_{int}	Internal-stress weight (DS only)	1.0
w_h	Momentum-slack penalty	800
h_{qp}	Spin-only safeguard band	$\pm 3.0 \text{ N}\cdot\text{m}\cdot\text{s}$
α_{torque}	Torque-minimisation weight	5
α_{reg}	Acceleration-regularisation weight	1.0
$T_{j,\max}$	Joint-torque limit	20 N·m
$f_{\text{qp}}^{\text{max}}$	QP contact-force box	3000 N
$\tau_{\text{qp}}^{\text{max}}$	QP contact-moment box	300 N·m
$\dot{q}_{\max}$	Joint-acceleration bound	50 rad/s ²
—	qpOASES working-set cap	500
<i>Attitude control (feedforward + PD)</i>		
K_θ	Attitude proportional gain	1.0 N·m/rad
K_ω	Rate damping gain	50.0 N·m·s/rad
K_d	Second-order damping gain	25.0 N·m·s ² /rad
K_{bw}	Momentum boundary-recovery gain	2.0 s ⁻¹
<i>System / scenario</i>		
—	Robot/structure mass ratio	1%
m_{robot}	Robot mass	71.06 kg
m_{struct}	Structure mass	7110 kg
$\mathbf{I}_{\text{struct}}$	Structure inertia	597/1493/1777
—	DoF per arm	7

at every single-support NMPC knot. The planner drives it *onto* the bound on two of the three axes (z on half the single-support samples, y on 7%, x never): the constraint is *active*, the binding limit shaping the plan. Fig. 6 shows what the plan does per step with and without it; Fig. 5 (top) shows that the resulting realised structure-on-robot moment also reaches the bound, on z , at the peak-demand step; the physical wheel momentum reported below is the empirical check on the reduced storage prediction. The reduced predicted-storage constraint, by contrast, remains inactive. The measured physical wheel momentum reaches a maximum per-axis magnitude of $4.10 \text{ N}\cdot\text{m}\cdot\text{s}$ on the binding axis (82% of the $5 \text{ N}\cdot\text{m}\cdot\text{s}$ allocated box, Fig. 5, middle). The resulting 18% reserve is a property of this operating point rather than of the rate constraint (Section VII-C); the growth of the physical stored peak with

TABLE 4. Docking performance over the six-step traversal, at the instant the weld engages. Capture requires a conjunction of three conditions: the gripper-to-anchor distance below 5 mm, the orientation error below 5° , and the length-normalised twist ξ_{cap} below 0.05 m/s ($\ell_c = 1$ m, Fig. 2). The twist columns give ξ_{cap} and its linear and angular blocks. The gate refused a pose-valid first approach on the twist condition twice on step 3 ($\xi_{\text{cap}} = 0.0605$ and 0.0574 m/s) and twice on step 5 (0.0576 and 0.0506 m/s); the rows above are the accepted captures.

Step	Arm	Pos. [mm]	Ori. [deg]	ξ_{cap} [m/s]	lin. [m/s]	ang. [rad/s]
1	B	4.02	0.14	0.0201	0.0148	0.0135
2	A	4.89	0.04	0.0077	0.0073	0.0026
3	B	4.99	0.02	0.0058	0.0054	0.0022
4	A	4.97	0.05	0.0072	0.0067	0.0025
5	B	4.95	0.01	0.0050	0.0047	0.0018
6	A	4.62	0.04	0.0070	0.0066	0.0025

Thresholds: 5 mm, 5° , 0.05 m/s on ξ_{cap}

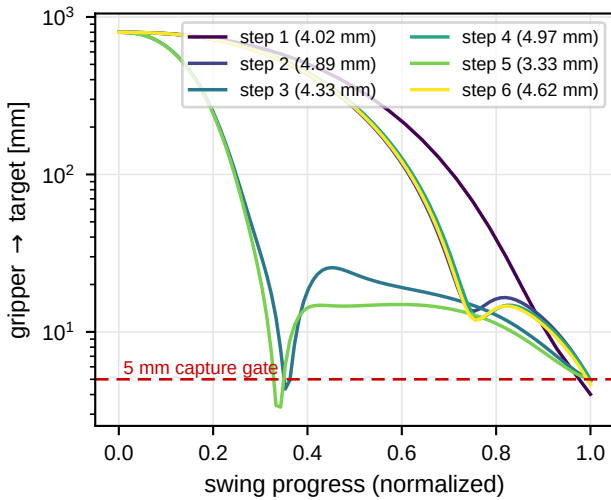


FIGURE 4. Swing end-effector approach to its target anchor, one curve per step on a normalised swing axis (log scale): each swing carries the gripper from the 0.8 m step length to the millimetre scale, and all six dock. The 5 mm line is only the distance part of the capture gate [19], which also gates ξ_{cap} : the closest approach can therefore be smaller than the at-weld distance (step 5 reaches 3.33 mm in flight but welds at 4.95 mm, Table 4).

crawl distance is analysed in Section VIII. The applied wheel command $\tau_{w,i}$ (Fig. 5, bottom) reaches its ± 2.5 N·m saturation in 95 of 2077 exported samples (4.57%). This quantity must not be read as a 100 Hz saturation duty: the exported table is phase dependent, with predominantly 0.01 s spacing in the inter-step double-support settle and 0.10 s spacing in much of single support. The phase-conditioned counts are $20/508 = 3.94\%$ in single support and $75/1368 = 5.48\%$ in the inter-step double-support settle, with no clipped terminal-hold sample. The predictive layer is bypassed in that inter-step double-support mode, so the aggregate exported-sample fraction cannot by itself be attributed causally to the Stage-1 constraint.

The distinction between the planner allocation and the physical AOCS command is also visible directly in the

canonical log. Before clipping, one command component reaches 3.899 N·m and the command-vector 2-norm reaches 4.583 N·m, although the planned robot-induced reaction is bounded by 2.5 N·m per axis. The aggregate feedback contribution (attitude, rate, acceleration-damping and boundary-recovery channels) reaches per-axis maxima of 0.606, 1.399, and 0.257 N·m. Thus (V-C.6) is not satisfied by construction in the reported configuration: its reserve is set to zero. The boundary-recovery channel itself is inactive because $\mathbf{h}_w$ never leaves the allocated box. These data support the narrower claim used throughout this paper: Stage 1 constrains the *allocated robot-induced reaction*; it does not constitute a hard bound on the sum of feedforward and feedback AOCS torques. A deployment-oriented implementation should choose a nonzero $\tau_{\text{fb,res}}$ from a robust feedback bound or a validated high-quantile envelope and should enforce the same budget through the support transitions.

Effect of the constraint. To isolate what the hard constraint contributes, the scenario was re-run in three configurations differing only in the constraint set given to the centroidal NMPC: neither bound, the rate bound alone, and both, the last being the canonical run. Everything else was held fixed, including the whole-body QP's envelope box and the attitude controller's output clip: holding the executed-side envelope fixed makes the comparison a statement about the plan rather than about the actuator.

Unconstrained, the planner drives the binding-axis reaction torque to 12.74 N·m on the long-reach swings, $5.1\times$ the envelope (Fig. 6), precisely where the managed plan pins at 2.500 N·m. The per-step maximum reaches the bound on all six steps: even the short-reach swings demand up to 3.8 N·m unmanaged, so no step leaves the constraint slack.

The raw exported-sample clip fraction is $203/2000 = 10.15\%$ in the no-envelope run and $95/2077 = 4.57\%$ in the canonical run, but this aggregate number is confounded by the phase-dependent logging cadence. More importantly, the direction of the difference is not uniform across phases: in single support the clipped-sample fraction is $14/422 = 3.32\%$ without the Stage-1 envelope and $20/508 = 3.94\%$ with it, whereas in the inter-step double-support settle it is $189/1377 = 13.73\%$ without the Stage-1 envelope and $75/1368 = 5.48\%$ with it. The latter phase is precisely the phase in which the predictive layer is bypassed. Consequently the $10.15\% \rightarrow 4.57\%$ exported-sample reduction is reported as a descriptive closed-loop observation, not as causal evidence that the NMPC reduces physical actuator saturation.

A time-weighted proxy leads to the same caution. If each exported command is held until the next timestamp, the traversal-window saturation proxy is 5.88% without the Stage-1 envelope and 4.26% in the canonical run. This reconstruction is still not a true uniformly logged 100 Hz duty and is not used as a performance claim. Table 5 reports both the phase-conditioned sample counts and this proxy so that the sampling dependence is explicit.

What the ablation does establish directly is a planning-layer effect. Unconstrained, the Stage-1 optimum requests

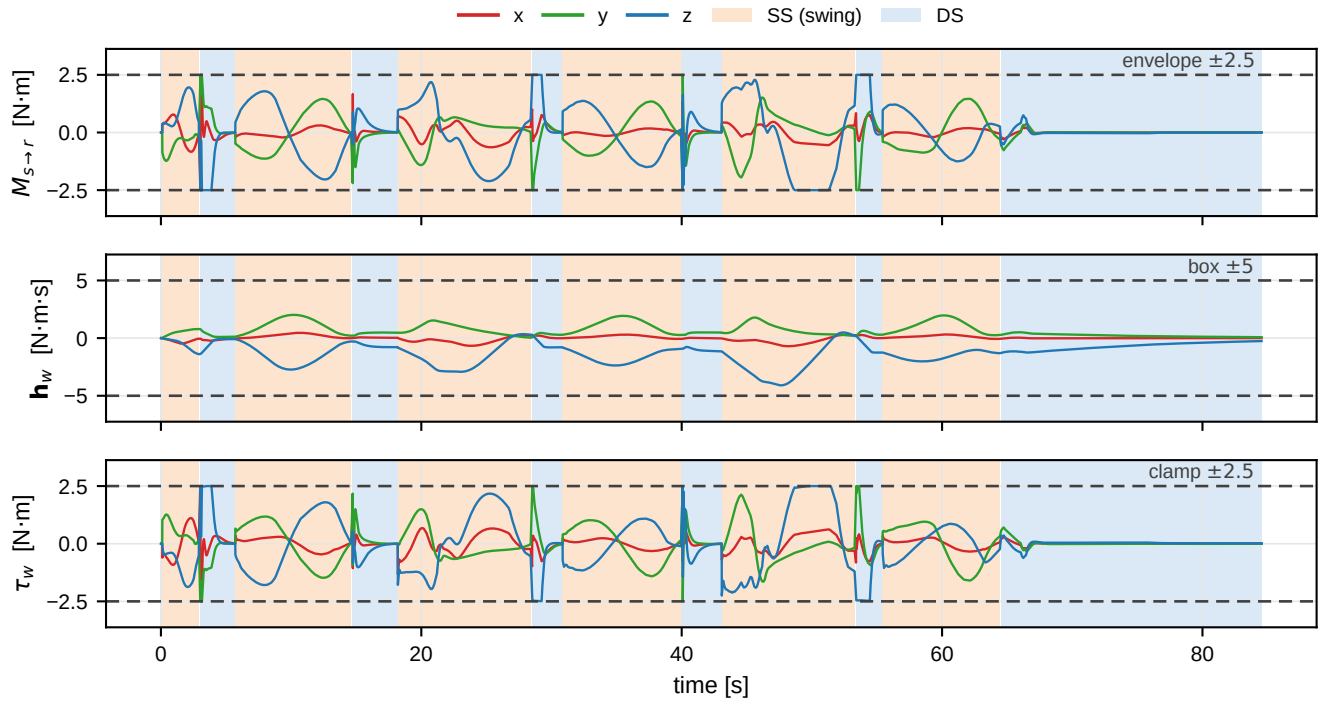


FIGURE 5. Actuation over the whole run, per axis in R_s ; shaded bands are single-support phases. *Top*: realised structure-on-robot moment $M_{s \rightarrow r}$ (V-B.2) against the ± 2.5 N-m envelope (dashed); the robot-on-structure reaction $\tau_{r \rightarrow s}$ is its negative, constrained identically by the symmetric bound, and the envelope is reached on z at the peak-demand step. *Middle*: stored wheel momentum h_w against the ± 5 N-m-s allocated box; the binding axis peaks at 4.10 N-m-s (82%) and returns toward zero between steps. *Bottom*: applied wheel torque τ_w against the ± 2.5 N-m physical clamp. The clamp is active in 95/2077 exported samples; because the export cadence is phase dependent, this count is a sample statistic rather than a uniformly sampled saturation duty.

TABLE 5. Saturation statistics from the retained exported logs. SS and DS are clipped exported samples divided by exported samples in that phase. “Time proxy” is the traversal-window fraction obtained by holding each exported command until the next timestamp; it is not a uniformly sampled 100 Hz duty.

Run	SS	DS	Time proxy
Canonical	20/508 (3.94%)	75/1368 (5.48%)	4.26%
No envelope	14/422 (3.32%)	189/1377 (13.73%)	5.88%

up to 12.74 N-m, whereas the constrained optimum remains inside the 2.5 N-m allocated reaction box. The whole-body QP then bounds the realised requested wrench in both cases, which explains why the traversal attitude peak is essentially unchanged: 0.3664° without the Stage-1 envelope and 0.3668° with it. Lifting the QP box as well, together with the attitude-controller clip, raises the peak to 0.89° ; because two executed-side mechanisms are removed together, that retained comparison is diagnostic rather than a factorial attribution.

At this operating point the rate-constrained run also exhibits the *higher* peak storage use, 82% against 70% of the box; no causal impulse decomposition is inferred from that difference. Adding the storage bound to the rate bound changes nothing measurable: the two configurations dock at the same instants and distances, reach the same attitude

and stored-momentum peaks, and differ by 0.4% in solver iterations. The storage inequality is present but never active. Accordingly, the present data do not demonstrate a finite-horizon advantage over an otherwise equivalent horizon-free controller; a storage-active operating point and an $N = 1$ baseline are required for that identification (Section VIII). Docking succeeds in all three configurations. The experimentally supported role of Stage 1 here is therefore to shape the *planned* robot-induced reaction before the executing QP, not to establish that the total physical actuator command remains unsaturated.

The envelope is not free: the managed locomotion reaches its final weld at 64.5 s against 56.0 s unmanaged (84.5 against 76.0 s logged, terminal hold included). The 8.5 s traversal difference is concentrated in step 3, whose single-support duration increases by 1.5 s, and step 5, which increases by 7.0 s; each other step differs by at most 0.2 s. In the managed run, the twist gate rejects two pose-valid evaluations on step 3 ($\xi_{\text{cap}} = 0.0605$ and 0.0574 m/s) and two on step 5 (0.0576 and 0.0506 m/s). The closest rejection misses the bound by 1.2%. These per-step comparisons show that the duration penalty is coupled to the capture criterion and is not an intrinsic property of the envelope alone. A cleaner measure of the same price is computational, and it depends on neither hardware nor capture threshold. The rate constraint costs the interior-point solver 10.83 iterations per solve on

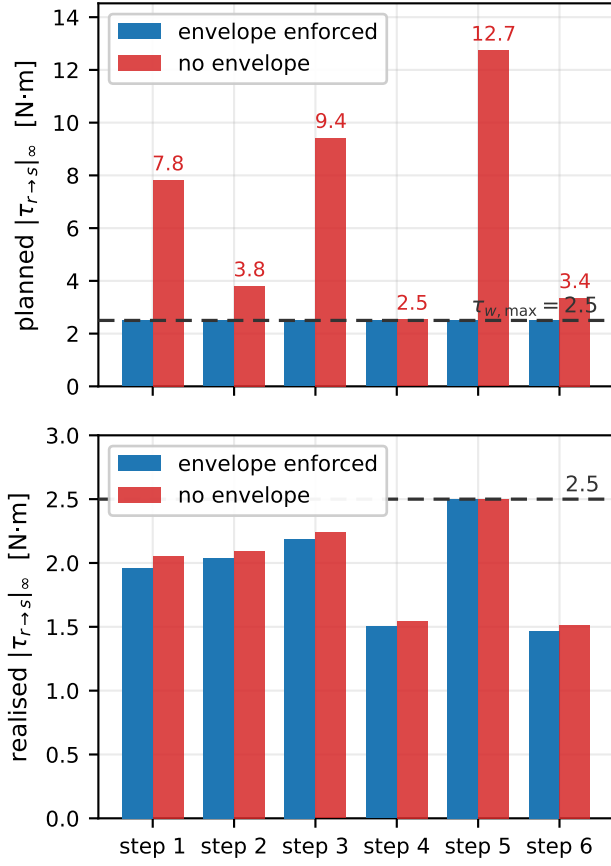


FIGURE 6. Effect of the envelope constraint on the plan and on the realised system, with the whole-body QP envelope and attitude clip held fixed. *Top:* per-axis reaction demand planned by the NMPC per step – enforced (blue), the plan rides the 2.5 N·m bound on every step; unenforced (red), it reaches 12.74 N·m ($5.1 \times$ the envelope) on the longest swing and 2.5–3.8 N·m on the short-reach steps. *Bottom:* realised reaction magnitude $\max_{t \in \text{step}} \|\tau_{r \rightarrow s}(t)\|_\infty$; the QP carries the same envelope on the requested wrench (VI-D.8), so the excessive unconstrained request is not realised. A rate-only run is omitted: indistinguishable from the enforced one at this operating point.

average against 6.91 with no bound, an increase of 57%, while the storage bound adds a further 0.4% (Section VII-G). Both figures reproduce exactly on replay.

D. PLATFORM ATTITUDE

Fig. 7 reports the platform attitude as the vector $\theta_s = \log(\mathbf{R}_s(0)^\top \mathbf{R}_s(t))^\vee$, one curve per component; the scalar geodesic angle is its norm. Per-axis figures, including the 0.37° traversal peak, are maxima of $|\theta_{s,a}|$. With the attitude controller active throughout, the attitude follows a charge-decay cycle: each component grows during the swings, as the wheels absorb the robot-induced momentum, and is drawn back down during the inter-step settles. No axis exceeds 0.37° during the traversal, and the settled value at the end of the inter-step double supports has a median of 0.10° . The controller works hardest immediately after each dock, where it rejects the contact-engagement transient; no attitude component changes by more than 0.11° over an entire inter-

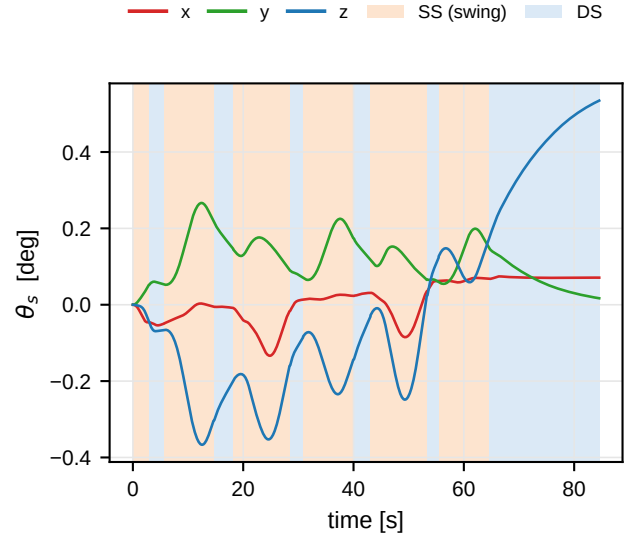


FIGURE 7. Platform attitude over the traversal, per axis in R_s . With the wheels commanded at every cycle, the attitude follows a charge-decay pattern, peaking at 0.37° per axis, well within an illustrative 5° pointing reference; no component changes by more than 0.11° over an inter-step double-support window. Shaded bands are single-support phases. The rise after the final weld is the structure absorbing the residual wheel momentum ($h_{w,z} = -1.28$ N·m·s); Fig. 8 follows its full recovery.

step double-support window.

E. POST-DOCK STATION KEEPING

The station-keeping extension follows the post-weld attitude rise of Fig. 7 to rest (Fig. 8). At the final weld the wheels store $h_{w,z} = -1.28$ N·m·s, the residue of rejecting the last swing. With the robot at rest and the total momentum conserved at zero, the structure must counter-rotate at $\omega_s = -h_w/I_{\text{comp}}$ while the wheels are loaded; the measured rate reproduces this pointwise (median ratio 1.01, implied composite inertia within 1.3% of the model). The attitude climbs while the feedback arrests the spin, peaking at $\theta_{s,z} = 0.59^\circ$ some 40 s after the weld, and the two rate states, locked by conservation, close together: $\|\mathbf{h}_w\|_\infty < 0.05$ N·m·s from 164 s after the weld and $\|\omega_s\| < 10^{-3}$ deg/s from 257 s. The accumulated angle then bleeds off exponentially with $\tau = 299$ s, crossing 0.05° at 796 s after the weld; a quasi-static residual of 0.048° remains on the y axis, 1% of that reference, at the end of the window. That residual is the expected steady state of a proportional law with no integral term: a persistent couple below a milli-newton-metre, exerted by the welded double support through the longest available lever (2.4 m about y), parks that axis at exactly this offset.

The conservation identity holds to $\|\mathbf{L}_{\text{tot}}\| \leq 1.5 \times 10^{-3}$ N·m·s over the whole run; the residual is acquired in discrete increments at the docking events, where weld activation introduces a small numerical total-momentum residual in the constraint solver, and is constant to the last exported digit over the 844 s of trace after $t = 120$ s (terminal hold 900 s), so it cannot produce a growing attitude error. It sits

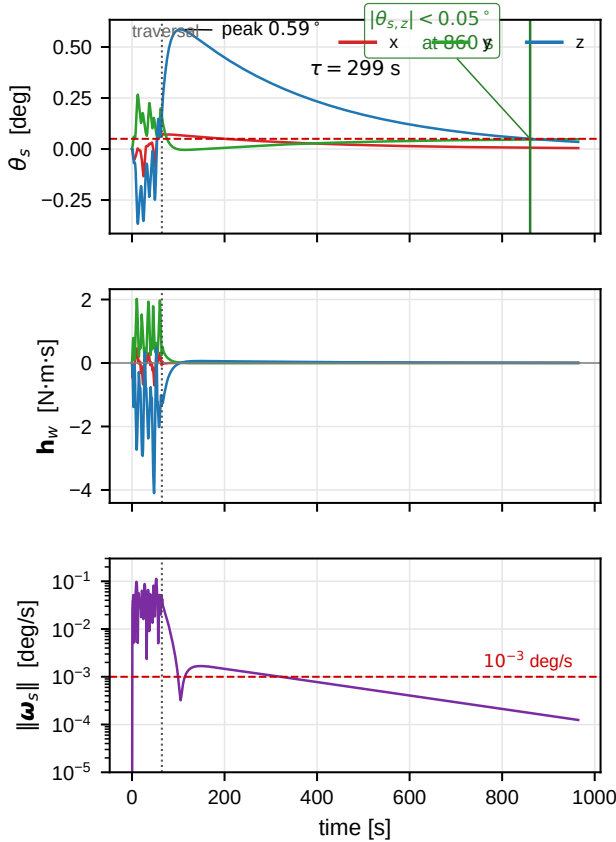


FIGURE 8. Station-keeping extension of the managed run (terminal hold 20 → 900 s, traversal bit-identical). *Top:* attitude per axis; the z component peaks at 0.59° while the wheels arrest the conservation-required spin, decays with $\tau = 299$ s and crosses 0.05° at 796 s after the weld (marker); a 0.048° quasi-static residual remains on y . *Middle:* wheel momentum per axis, the -1.28 N·m·s arrest load unwinding. *Bottom:* structure rate (log scale); the notch near $t = 105$ s is the spin reversing. Locomotion occupies the first 64.5 s (dotted line).

at the same order as the end-of-window wheel momentum (4×10^{-3} N·m·s), so the tail's last digits are indicative; the peak excursion and the 299 s constant rest on quantities an order of magnitude larger.

The reduced prediction that the storage constraint acts on can be checked one step ahead against the measured wheel momentum. Over the 703 consecutive NMPC-active pairs the one-step error has an infinity-norm median of 0.049 N·m·s and a maximum of 0.286, at worst 5.7% of the allocated box. This validates only the first lead time. The solved $\{\hat{\mathbf{h}}_{w,k+i|k}\}_{i=1}^N$ trajectory was not persisted in the retained run, so neither horizon-end prediction error nor a lead-time error envelope can be reconstructed. We therefore make no full-horizon prediction-accuracy claim from this data set.

Through the recovery the commanded wheel torque stays in the milli-newton-metre range (10 mN·m at the attitude peak, below 1 mN·m in the tail): the locomotion-tuned feedback, with unchanged law and gains, acting on a quiet structure.

F. WHOLE-BODY TRACKING

Figs. 9 and 4 show that the controller executes the planned motion. During single support the whole-body QP holds the torso orientation to an RMS of 0.83° (peak 1.68°), while the swing end-effector is driven from the 0.8 m step length down to the docking tolerance, its orientation converging to the target by capture.

Two quantities describe the torso along the progression axis. The *physical recoil*: as the swing arm accelerates, momentum conservation pushes the base backwards by ≈ 25 mm on the long-reach swings (24, 27 and 26 mm; 3% of the 0.8 m stride). The *tracking error* against the planned quintic of Appendix A peaks higher — 164 mm at mid-swing on the long-reach steps, 44–51 mm on the short-reach ones, single-support median 32 mm — because the geometric quintic ignores the momentum coupling and advances while the reaction pushes the torso back. The error is transient: the torso reconverges to within 18–27 mm by the end of each inter-step hold (99 mm on the initial step, which departs from the deployment pose). The centre of mass, regulated at the momentum level by the CoM linear-momentum task, tracks more loosely (Fig. 9, bottom): a single-support median of 61.5 mm in norm, of which 47 mm along the progression axis, about 2% of the 2.2 m that axis travels (Section VIII). In the inter-step holds the just-landed arm remains seated at its anchor to the micrometre level, confirming the contact constraint throughout the transition.

G. COMPUTATIONAL COST

Iteration counts are deterministic and reproduce exactly on replay; wall-clock timings are hardware-dependent and move by roughly a quarter between machines, so the timings are quoted on a stated machine. Of the 2077 logged control ticks, 709 carry an NMPC solve (508 single-support and 201 terminal-hold) and the remaining 1368 inter-step ticks bypass the predictive layer; the whole-body QP is scheduled ten times per NMPC tick and once per tick otherwise, giving $1368 + 709 \times 10 = 8458$ solves. No deadline-miss handling is implemented: the simulation is not run in real time and a late solve simply takes longer. The NMPC converges in a median of 11 interior-point iterations (mean 10.87, 95th percentile 15, maximum 18), and every solve returns a success status: 704 `Solve_Succeeded` and 5 `Solved_To_Acceptable_Level`, with no failures and no restoration phases. The whole-body QP completes its 8458 solves at a median of 90 working-set recalculations; the backend reports success on all of them. A coarse pre-planner solves one nonlinear program per step (six in total, all converged; median 13 iterations, 40 ms); it runs once before each step, outside the online loop, and is excluded from the per-tick budget.

Timings are measured on a four-core Intel Xeon at 2.10 GHz under Linux, on a replay bit-identical in its deterministic channels; the implementation is unoptimised pure-Python research code, and the measured loop carries the simulator's own integration and logging.

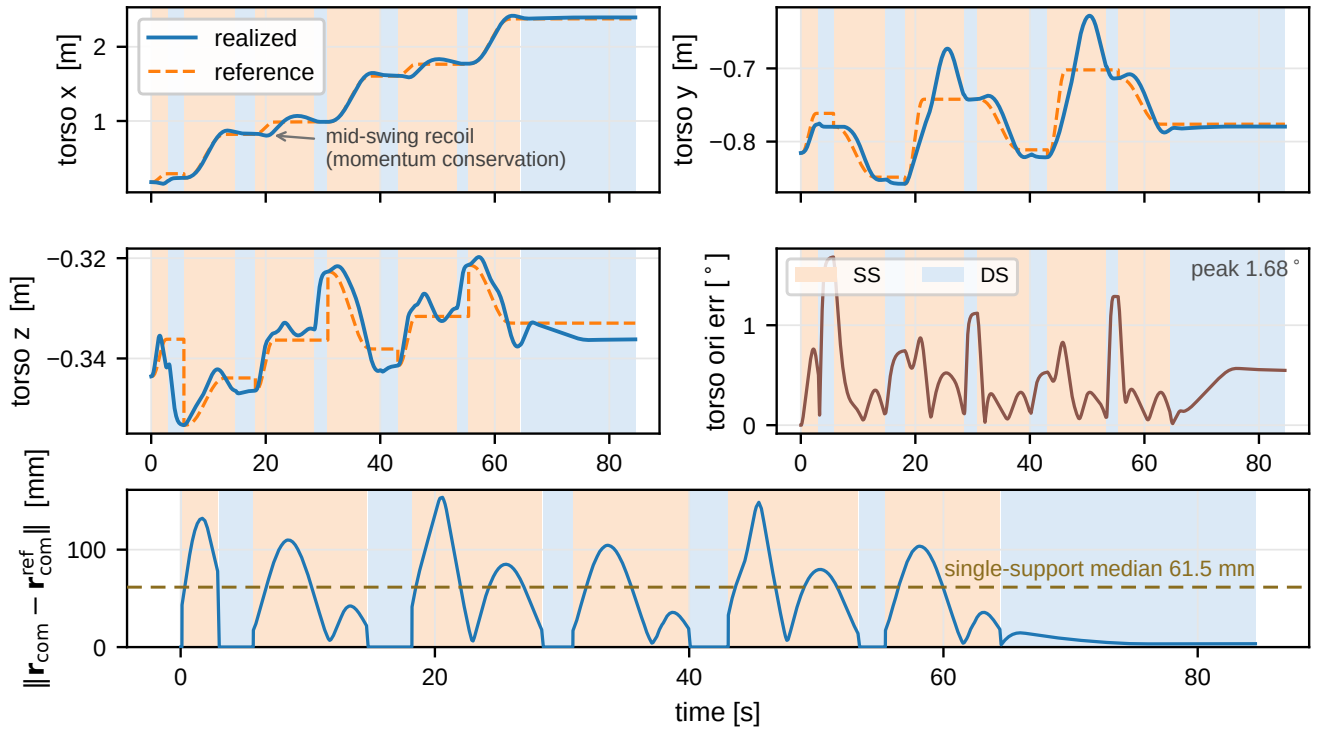


FIGURE 9. Torso and CoM tracking in the rotating structure body frame R_s ; shaded bands are single-support phases. *Top rows:* torso position per axis (solid: realised, dashed: reference) and orientation error to the constant reference R_{flat} (RMS 0.83° , peak 1.68°); the mid-swing excursion is analysed in Section VII-F. *Bottom:* CoM tracking-error norm to the NMPC reference (single-support median 61.5 mm in norm, 47 mm along the progression axis).

The NMPC solver takes a median 20.7 ms and a maximum of 37.1 ms over its 709 solves; its solver-only equivalent rate is therefore at least 27 Hz over this replay, above the 10 Hz invocation rate. The QP is profiled per exported tick: each of the 2077 records stores the mean and the worst of its one to ten solves. The per-tick mean solve time has a median of 5.6 ms and a 99th percentile of 8.9 ms. The QP accounts for about 71% of the measured whole-body block, the remainder being multibody evaluation, attitude control and simulation.

Counting the worst QP solve in each exported tick, a solve exceeds 10 ms on 22 of the 2077 ticks (1.1%), and on 3 of the 508 single-support ticks (0.6%). Eighteen of the twenty-two occur in the terminal settle. One isolated solve reaches 98 ms at the transition into that settle; the plausible cause, construction of a solver instance for a new problem shape, is unverified, so the outlier is retained. These are component-level solver timings, not an end-to-end real-time validation: the 1.1% statistic refers only to exported ticks whose worst profiled QP solve exceeds 10 ms.

H. OBSERVED LIMITATIONS

Three points are noted for completeness. First, the CoM position lags its reference during dynamic swings (single-support norm median 61.5 mm, Section VII-F); this does not affect docking and motivates the path-following reformulation of Section VIII. Second, the planner constraints are knot-wise single-support statements. Between knots and in the inter-step

double-support settle they are not enforced by the predictive layer; the QP continues to limit its requested wrench, while the physical AOCS command can still clip because feedback lies outside the Stage-1 budget. Third, neither recursive feasibility nor robustness beyond the frozen 1% deterministic configuration is established. A by-construction full-cycle envelope, explicit feedback reserve and uncertainty campaign are therefore separate validation requirements rather than consequences of the reported nominal run.

VIII. LIMITATIONS AND FUTURE WORK

The reported results establish a planning-layer actuation allocation under a specific rigid-body, deterministic operating point; they do not establish a hardware-independent closed-loop guarantee. The main limitations are therefore stated here in the same variables as the claimed result.

Actuator model and feedback reserve. The plant uses three orthogonal, fixed-axis reaction wheels. For that abstraction the independent per-axis storage box is well defined. A control-moment-gyroscope cluster, by contrast, has a directional torque/momentum set that depends on gimbal geometry and can contain internal singularities [23], [31]; a Cartesian box is admissible only after it has been shown to lie inside the cluster's joint steering-feasible set [32], [33]. No such CMG validation is claimed here. Even for the reaction-wheel plant, the Stage-1 bound applies to the allocated robot-induced reaction, not to the total AOCS motor

command. The canonical configuration uses zero feedback reserve, and the retained log contains pre-clipping commands above the 2.5 N·m per-axis physical clamp (Section VII-C). A hardware-facing implementation should therefore enforce Eq. (V-C.6) with a nonzero $\tau_{fb,res}$, obtained from a robust bound or validated uncertainty envelope for the feedback channels.

Finite-horizon contribution and storage activity. Of the two Stage-1 path constraints, only the planned reaction-torque bound becomes active in the reported traversal. The reduced storage inequality remains inactive, and the rate bound is pointwise in the contact wrench. Consequently these data do not identify an advantage of $N = 8$ over an otherwise equivalent horizon-free or $N = 1$ controller. This is an experimental identifiability limitation, not a mathematical absence of the storage mechanism: because $\mathbf{h}_w(t) = \mathbf{h}_w(0) + \int_0^t \boldsymbol{\tau}_m(s) ds$, storage is an integral state for which future reaction allocation can matter. The decisive validation is a storage-active operating point with all other controller settings frozen, compared at least for $N \in \{1, 2, 4, 8\}$. Such a campaign must report the minimum storage margin, docking completion, traversal time and control effort rather than selecting only a successful horizon after the fact. Until that experiment is performed, the present paper deliberately makes no claim that a finite prediction horizon is necessary at the canonical point.

Hybrid-cycle coverage. The NMPC is bypassed during the inter-step double-support settle. This matters experimentally: most clipped exported samples in the canonical run occur in that phase (Section VII-C). The whole-body QP remains active and limits its requested contact-wrench reaction, but the Stage-1 allocation is not a by-construction SS→DS→SS guarantee. A direct extension is a contact-scheduled hybrid NMPC in which the contact set changes from one to two active wrenches while the same reaction and storage budgets remain enforced. The same extension should retain a nonzero AOCs feedback reserve.

Knot-wise and recursive feasibility scope. The path constraints are imposed at the transcription nodes of a feasible solve. They do not by themselves exclude an inter-sample excursion, and no terminal invariant set or other proof of recursive feasibility is included. A stronger implementation would either enforce the reaction/storage constraints at the Runge–Kutta substeps (or by collocation) or validate every accepted plan on a finer temporal mesh. Recursive feasibility would require a terminal admissible set or a separate viability argument. The reported solver produced a usable solution at every NMPC update, but no online infeasibility fallback was exercised; a flight-oriented implementation should retain the last feasible plan or enter a constraint-preserving hold rather than fall back to an unconstrained command.

Prediction accuracy. Only the first predicted storage step can be reconstructed from the retained logs. Its infinity-norm error has median 0.049 N·m·s and maximum 0.286 N·m·s, but the full predicted trajectory was not persisted, so the error growth with lead time is unknown. Future runs should store $\{\hat{\mathbf{h}}_{w,k+i|k}\}_{i=1}^N$ and report median, 95th-percentile and worst-

case error versus i . If the resulting envelope is non-negligible, a constraint tightening $\|\hat{\mathbf{h}}_{w,k+i|k}\|_\infty \leq h_{\max} - \delta_h(i)$ should replace the nominal storage bound.

Dynamics and uncertainty. The structure is rigid and the docking event is represented by a weld constraint. Flexible modes, compliant capture, estimation error, command/measurement latency and actuator-model mismatch are not included. The present 1% robot-to-structure mass ratio is also a single operating point; higher coupling reduces the available margin and the current untuned gait does not complete the same traversal. Robustness should therefore be evaluated independently in robot mass and host inertia, not by mass ratio alone, and should include initial wheel loading, telemetry noise, latency and contact uncertainty. A finite-stiffness capture law is likewise needed to test whether the conclusions survive a non-instantaneous docking transient.

Reference parametrisation. The CoM linear-momentum task governs the centre of mass at the rate level, and its position lags the reference during dynamic swings (single-support norm median 61.5 mm). Because the admissible transverse motion tightens as the lever arm grows, a fixed time-parametrised trajectory is increasingly ill matched to the feasible set. A geometric path $\mathbf{r}(s)$ with path rate $\dot{s} \geq 0$ as an optimisation variable would allow the planner to slow where the reaction allocation becomes restrictive, and would separate geometric task completion from arbitrary timing.

The lever-arm trend observed in the canonical traversal is physically consistent with the orbital term $\mathbf{r}_{\text{com}} \times \sum_j \mathbf{f}_j$ in Eq. (V-B.2c): over three long-reach swings the binding-axis stored peak rises from 1.40 to 4.10 N·m·s as $\|\mathbf{r}_{\text{com}}\|$ grows from 0.69 to 1.76 m. The corresponding linear fit ($R^2 = 0.95$) is only an indicative trend from one traversal, not a validated scaling law. Reactionless or angular-momentum-neutral swing planning in the base-coupling null space [2] is complementary to the allocation approach considered here.

IX. CONCLUSION

We presented an actuation-envelope-constrained motion-generation and control architecture for a dual-arm robot crawling on a free-floating structure. A centroidal NMPC generates online references under a per-axis bound on the *planned robot-induced reaction* and a locally refreshed reduced prediction of wheel storage, both obtained from conservation about the structure centre of mass. A whole-body QP then executes the references under the multibody and contact constraints, while the reaction-wheel attitude controller remains a separate subsystem; its state is not co-integrated in the locomotion dynamics.

For the reported six-step traversal at a 1% robot-to-structure mass ratio, all six docking transitions satisfy the capture gate. The Stage-1 reaction constraint is active and limits the planned demand to 2.5 N·m per axis, whereas the same optimiser without the envelope requests up to 12.74 N·m. The executing QP prevents that excessive request from being realised, which is why the constrained and unconstrained Stage-1 runs have nearly identical sub-degree attitude peaks.

The physical wheel momentum reaches 4.10 N·m·s, leaving 18% margin to the allocated storage box; the storage constraint itself never becomes active. The retained data therefore support reaction-demand shaping at this operating point, but not a claim that a finite horizon is necessary for storage management.

The same distinction applies to actuator saturation. The planner constrains the allocated robot-induced reaction, whereas the physical AOCS command also contains feedback. With zero feedback reserve, its pre-clipping command can exceed the 2.5 N·m clamp, and the mixed-rate exported clip statistic cannot be interpreted as a causal Stage-1 benefit. A complete hardware-facing validation must add a nonzero feedback reserve, enforce the envelope through the double-support transitions, record the AOCS command uniformly, and compare a storage-active case against an $N = 1$ baseline. Those experiments define the remaining step from the demonstrated planning-layer envelope to a full-cycle predictive momentum-management claim.

APPENDIX A REFERENCE GENERATION

This appendix gives the geometric constructions producing the two 6-D references Stage 2 tracks directly: the torso reference, generated per step from the stance-arm anchor positions, and the swing reference.

a: Position – quintic polynomial.

The torso displacement target is a fraction $\kappa_f = 0.70$ of the full IK-derived displacement $\Delta \mathbf{r}$, starting after a delay $\kappa_d = 0.20$ of the swing duration T_{sw} . With $t_s = T_{ss} + \kappa_d T_{sw}$ and $t_e = T_{ss} + T_{sw}$ (T_{ss} the single-support start time), a zero-boundary quintic $s(t) \in [0, 1]$ provides C^2 profiles:

$$\mathbf{r}_t(t) = \mathbf{r}_{t,0} + s(t) \kappa_f \Delta \mathbf{r}, \quad t \in [t_s, t_e], \quad (\text{A.1})$$

where $\mathbf{r}_t$ is the torso position of the generalised coordinates (Section IV-A) and $\mathbf{r}_{t,0}$ its value at swing start. The quintic (A.1) is tracked *directly* by the torso-pose task of Section VI-C; it is a function of the anchor geometry alone.

b: Orientation – constant globally-held reference.

The torso orientation reference is *not* re-anchored per phase. A single rotation $\mathbf{R}_{\text{flat}} = \mathbf{R}_t(0)$, the torso attitude at mission start (the mounting orientation), is stored once and held as the orientation target for the entire traversal, across every SS and DS phase:

$$\mathbf{R}_t(t) \equiv \mathbf{R}_{\text{flat}}, \quad \dot{\omega}_{\text{torso}}(t) = \mathbf{0}, \quad \forall t, \quad (\text{A.2})$$

so the planned centroidal-angular-momentum reference passed to the NMPC is simply $\hat{\mathbf{L}}_{\text{com}} = \mathbf{0}$. The crawling gait progresses by *translation*: the near-term sensor payload (stereo vision) requires a fixed pointing. Because the target never depends on the finite-state machine, no phase transition can introduce a reference jump, and the torso returns to $\mathbf{R}_{\text{flat}}$ after each swing with a non-accumulating residual drift (Section VII).

c: Swing reference – quintic with clearance, delayed rotation.

Unlike the torso, the swing gripper must reach its anchor exactly. Its position follows the same zero-boundary quintic $s(\tau)$, $\tau \in [0, 1]$, applied to the *full* start-to-anchor displacement with no start delay, plus an out-of-plane clearance arc that lifts the gripper off the structure at mid-swing:

$$\mathbf{p}_{ee}(\tau) = \mathbf{p}_{ee,0} + s(\tau) (\mathbf{p}_a - \mathbf{p}_{ee,0}) + d_c \sin^2(\pi\tau) \mathbf{n}, \quad (\text{A.3})$$

where $\mathbf{p}_a$ is the target anchor, $\mathbf{n}$ the outward structure normal and $d_c = 0.03$ m the clearance. The orientation rotates from the release orientation toward a constant target on a delayed-cosine profile that stays still for the first 20% of the swing, so the arm clears the structure before turning. That target is the structure-frame identity, coincident with the axis-aligned anchor frames of this scenario.

DATA AND CODE AVAILABILITY

The controller implementation is publicly available at github.com/OcelotIC/CrawlBot_control. The canonical controller configuration used by the reported 2.5 N·m campaign was frozen at commit 32aefaf; the submission package contains the retained simulation tables and the scripts used to regenerate the manuscript figures. Because the public development branch may evolve, numerical values in this article are tied to the frozen submission artifacts rather than to the current repository head.

ACKNOWLEDGMENT

In accordance with the IEEE policy on the use of generative artificial intelligence, the authors disclose that Claude (Anthropic PBC), including the Claude Code agentic coding tool, was used to draft and revise text in all sections under the authors' direction; to write and debug the software tooling of the simulation study (data export, diagnostic analysis, figure generation and rendering); and to cross-check the numerical values reported in the text against the simulation artifacts. The problem formulation, control architecture, methodology, experiments, and conclusions are the authors' own; all AI-assisted output was reviewed and validated by the authors against the committed simulation data, and the authors take full responsibility for the content of this article.

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